INVESTING & MARKETS (INDIA)

Futures & Options Demystified

Derivatives without the hype — what SEBI's own studies say about retail F&O outcomes, futures and the cost of carry, calls and puts with worked payoff tables, implied volatility and the Greeks explained with numbers, thirteen strategies with real break-evens, SPAN margin and the penalty regime, the STT-on-exercise trap, physical settlement, the arithmetic of why 9 in 10 lose, the 2024–25 SEBI framework changes, and the handful of legitimate uses worked end to end.

derivatives · futures · options · nse · sebi · india · f&o · greeks · implied volatility · payoff diagram · span margin · option strategies · hedging · covered call · iron condor · risk

Study Pack · Comprehensive reference & interview prep · ~144 min read · 31,752 words

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Read this first: what actually happens to retail F&O traders

Most guides to futures and options open with a payoff diagram. This one opens with the outcome data, because the outcome data is the single most important fact about this market and almost nobody puts it first. If you read only one section of this guide, read this one.

This is financial education, not financial advice

Nothing here is a recommendation to buy, sell or trade any security, index, contract or strategy. Derivatives are leveraged instruments and you can lose more than you deposit when you are short options or long futures. Every number, rate, contract specification, margin rule and regulation below is illustrative and dated **as at late 2026** — rules in Indian derivatives have changed repeatedly and materially since 2024. Verify everything current at **nseindia.com**, **bseindia.com** and **sebi.gov.in** before you act. For personal advice, consult a SEBI-registered Investment Adviser (RIA). Past performance of any strategy tells you nothing reliable about its future.

The evidence

The Securities and Exchange Board of India has studied the profit and loss of individual participants in the equity derivatives segment, using actual exchange and broker data rather than surveys. Two studies matter.

Study	Published	Period covered	Headline finding (as reported)
SEBI study on individual traders in the equity F&O segment	September 2024	FY 2021–22 to FY 2023–24	Roughly **93%** of individual F&O traders incurred losses. Aggregate net losses for individuals across the three years were reported in the region of **₹1.8 lakh crore** (₹1,80,000 crore, i.e. about ₹1.8 trillion).
SEBI follow-up study on index options	2025	FY 2024–25	A broadly similar picture — roughly **91%** of individual traders in index options lost money, despite the curbs introduced from late 2024.

⚠ **Treat these as directional, not immutable**

SEBI has updated these studies and will update them again. The precise percentages, the aggregate rupee figures and the definitions of "individual trader", "net loss" and "transaction costs" vary between releases. Read the current primary source at **sebi.gov.in** rather than quoting a number you saw in a headline. What is *stable* across every version is the shape: a large majority lose, the losses are concentrated in small accounts, and transaction costs are a material share of the loss.

~93%
INDIVIDUALS LOSING, FY22–FY24 (SEBI, SEP 2024)

~91%
INDIVIDUALS LOSING IN INDEX OPTIONS, FY25 (SEBI, 2025)

~₹1.8 lakh cr
AGGREGATE INDIVIDUAL NET LOSS, FY22–FY24

0
GUARANTEED STRATEGIES IN THIS GUIDE

What that number means and what it does not mean

A 90%-plus loss rate does not mean the market is rigged, and it does not mean the 7–9% who profit are all geniuses. It means something more boring and more useful:

DERIVATIVES ARE CLOSE TO ZERO-SUM BEFORE COSTS

For every rupee a futures buyer gains, a futures seller loses exactly a rupee. For every rupee an option buyer gains, the writer loses it. There is no underlying economic growth being shared out the way there is with equity ownership. The pie does not grow; it only moves.

AND FIRMLY NEGATIVE-SUM AFTER COSTS

Brokerage, STT, exchange transaction charges, SEBI turnover fees, stamp duty, GST and the bid–ask spread all leak out of the pot before anybody settles up. Taken together those costs are a meaningful share of the money that individuals lose — SEBI's own analysis attributes a large chunk of small-trader losses to transaction costs alone.

THE COUNTERPARTIES ARE NOT AMATEURS

The consistent profit in Indian F&O accrues to proprietary trading firms, foreign portfolio investors running systematic books, and market makers. They have co-located servers, millisecond latency, negotiated cost structures, risk teams and inventory models. You are a retail participant with a mobile app. That is the matchup.

LEVERAGE COMPRESSES THE TIMELINE

With 5–10x effective leverage, an edge that would take a decade to show up in cash equities shows up — in either direction — in weeks. Leverage does not create skill. It accelerates whatever you already have, including the absence of skill.

Who actually makes money in this market

SEBI's studies break participants down by category. The consistent pattern across years is roughly this. Numbers below are illustrative of the shape reported, not exact; verify current figures.

Participant category	Typical net outcome	Why
Proprietary traders (prop desks)	Large aggregate net profit	Market making, statistical arbitrage, latency advantage, scale, negotiated costs, professional risk control.
Foreign portfolio investors (FPIs)	Large aggregate net profit	Algorithmic strategies, volatility arbitrage, cross-market hedging, institutional cost structures.
Domestic institutions / corporates	Mixed; often net hedgers	Using derivatives for their intended purpose — transferring risk, not seeking P&L.
Individuals (you)	Large aggregate net loss	No structural advantage; highest relative cost drag; behavioural errors; leverage.

The transfer is mechanical

Money flows from the net-loss category to the net-profit categories. When you buy a weekly index option two hours before expiry, there is a professional on the other side who has priced it, hedged it, and is happy to take the position at that price. The question is not "will the index move?" — it is "am I getting a better price than the firm that prices this for a living?"

Three legitimate reasons to use derivatives

Derivatives exist for good reasons. They were invented so that a farmer could lock in a price for a crop and a miller could lock in a cost for the flour. Retail participants have three defensible uses, and none of them is "make money fast".

1. **Hedging an existing exposure**

   You own ₹10,00,000 (ten lakh) of equity and you are worried about a drawdown over a specific window — a results season, a budget, an election. You can buy index put options or short index futures to offset part of that risk. You pay for the protection. You expect the hedge to lose money most of the time, because insurance that pays out most of the time is mispriced.

2. **Generating income against something you already own**

   You hold shares in a company you intend to keep. Writing a covered call against that holding converts some future upside into cash premium today. This is a genuine strategy with a genuine cost: you cap your upside and you still own all the downside. It is not free money and it is not "safe".

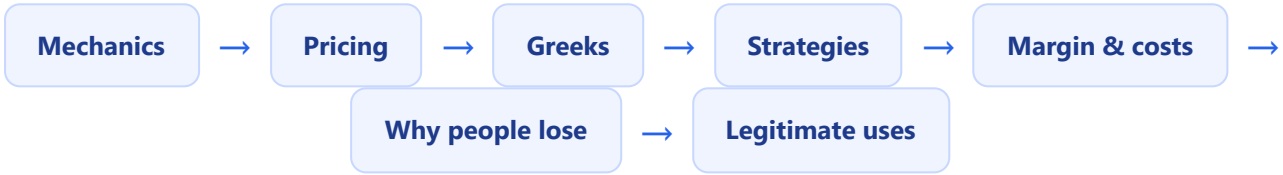
3. **Expressing a view with defined, capped risk**

You believe something specific will happen in a specific window. Buying a vertical spread or a single long option caps your loss at the premium paid. You know your maximum loss before you enter. This is the only speculative use that does not have an open-ended tail.

✕ **What is not on that list**

"Selling options for steady monthly income" with naked short positions. "Scalping Bank Nifty weeklies." "Buying cheap out-of-the-money calls on expiry day." Martingale-style averaging down on a losing option position. Any strategy whose description includes a claimed accuracy percentage. Each of these is covered later in this guide with the arithmetic showing why it fails.

How to read the rest of this guide



Everything that follows teaches mechanics, arithmetic and risk. Nothing that follows is a system. If you finish this guide and your conclusion is "now I know how to trade F&O profitably", you have misread it. The correct conclusion is "now I understand what this instrument is, what it costs, and under what narrow circumstances it is useful to me".

- ☐ You will not see a single named stock recommended anywhere in this guide.
- ☐ Every worked example uses a hypothetical index level (typically 25,000) and hypothetical strikes.
- ☐ Every contract spec and rate carries a "verify" note with a date and a source.
- ☐ Where a popular strategy is bad for retail, the arithmetic showing why is included.

What a derivative actually is

A derivative is a contract whose value is *derived* from something else. You are not buying the thing. You are buying a legally enforceable agreement about the thing — typically about its price at a future date. Everything else is detail.

The four components of any derivative

Component	What it is	Example (hypothetical)
Underlying	The asset or index the contract references	A benchmark equity index at a level of 25,000; or the shares of a listed company
Contract size / lot	How much underlying one contract controls	75 units of the index (lot size is set by the exchange and has changed repeatedly — verify at nseindia.com)
Expiry	The date the contract ceases to exist and is settled	The last Tuesday or Thursday of the month, depending on index and exchange — expiry days have been reshuffled; verify
Settlement	How obligations are discharged	Cash (index derivatives) or physical delivery of shares (single-stock derivatives in India)

Why leverage is the defining feature

Here is the arithmetic that makes derivatives dangerous, stated as plainly as possible. Assume a hypothetical index at 25,000 and a lot size of 75.

TEXT

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NOTIONAL EXPOSURE OF ONE INDEX FUTURES CONTRACT
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Index level                        25,000
Lot size                           75
Notional value   = 25,000 x 75 = Rs 18,75,000   (eighteen lakh seventy-five thousand)

Approximate initial margin (SPAN + exposure), say 12% of notional
Margin blocked   = 18,75,000 x 0.12 = Rs 2,25,000

Effective leverage = 18,75,000 / 2,25,000 = 8.3x

NOW MOVE THE INDEX BY 1%
────────────────────────

Index moves 25,000 → 25,250 (+250 points, +1.0%)
P&L on one lot = 250 x 75 = Rs 18,750 profit
Return on margin = 18,750 / 2,25,000 = +8.3%

Index moves 25,000 → 24,750 (−250 points, −1.0%)
P&L on one lot = Rs 18,750 LOSS
Return on margin = −8.3%

A 1% index move = an 8.3% move on your capital.
A 12% index move against you = your entire margin, and you owe more.

Margin percentages are illustrative. Actual SPAN + exposure margin
varies daily with volatility. Verify on the exchange margin calculator.
As at late 2026 − verify at nseindia.com.
```

Losses are not capped on futures or short options

If you are long futures, short futures, or short (written) options, your loss is not limited to the money in your account. A gap-down of 8% on a leveraged long position can leave you owing your broker money. Indian brokers will square you off, but a gap happens before any square-off can execute. This is not a theoretical risk; it has happened in every market, repeatedly.

Equity is positive-sum. Derivatives are not.

This is the most important conceptual difference and it is routinely glossed over.

BUYING EQUITY

You own a share of a business. The business generates earnings, reinvests, grows, pays dividends. Over long periods the total value of all shareholders' holdings grows because the underlying businesses produce real output. Every long-term shareholder can profit simultaneously. The pie grows.

TRADING DERIVATIVES

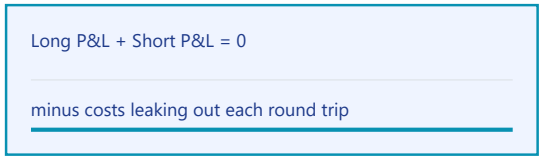
Every contract has a long and a short. Their P&L sums to exactly zero before costs. There is no earnings stream inside a futures contract. For you to make ₹50,000, somebody must lose ₹50,000. After brokerage, STT, exchange fees and spread, the two sides together lose money. The pie shrinks.

EQUITY OWNERSHIP (positive sum)



All holders can gain together over time.
Value created outside the market.

DERIVATIVES (zero sum, then negative)



Your gain is literally someone else's loss.
No value created; costs destroy value.

Consequence: in equity you can be mediocre and still do fine.

In derivatives you must be better than your counterparty, by more than the costs.

Hedgers are the exception: they accept negative expected P&L in exchange for reduced risk. That is rational.

Why "I do well in stocks so I should do well in options" does not follow.

The two families

	Futures (and forwards)	Options
Obligation	Both parties are **obliged** to settle at expiry	Buyer has a **right**, not an obligation. Seller has an obligation if the buyer exercises.
Upfront payment	No premium; margin only (returned when you close)	Buyer pays a premium (spent, not returned). Seller receives it and posts margin.
Buyer's max loss	Unlimited (down to zero underlying)	Limited to premium paid
Seller's max loss	Unlimited	Unlimited for calls; very large for puts
Payoff shape	Linear — straight line	Non-linear — kinked at the strike
Time decay	None (basis converges, which is different)	Yes — the premium erodes as expiry approaches

✦ **The single sentence summary**

A future is a *bet on direction with symmetric unlimited risk*. An option is a *transfer of risk for a price* — the buyer is paying for insurance or for a lottery ticket (same instrument, different intent), and the seller is the insurance company taking premium in exchange for an open-ended liability.

Vocabulary you need before the next section

Term	Meaning
Long	You bought it. You profit if it goes up (for a future or a call).
Short / written	You sold it without owning it. You profit if it goes down or stays put.
Notional value	Lot size × underlying price. Your real exposure, not the margin you paid.
Premium	The price of an option, quoted per unit of underlying. Multiply by lot size for the rupee amount.
Strike (K)	The fixed price written into an option contract.
Spot (S)	The current market price of the underlying.
Open interest (OI)	Number of contracts currently outstanding (not closed out). A measure of participation.
Mark to market (MTM)	Daily revaluation of positions; cash moves in and out of your account each day for futures.
Square off	Close an existing position by taking the opposite trade.
Expiry	The date the contract dies and settles.

Forwards, futures and the Indian futures market

A forward is a private bilateral agreement. A future is the same economic idea, standardised, listed on an exchange and guaranteed by a clearing corporation. That standardisation and guarantee is the entire innovation, and it is what makes exchange-traded derivatives usable by someone with a demat account rather than a treasury department.

Forward versus future, side by side

Dimension	Forward	Future
Where it trades	Over the counter, negotiated directly between two parties	On an exchange (NSE, BSE) with a visible order book
Terms	Fully customised — any quantity, any date, any underlying	Standardised — fixed lot size, fixed expiry dates, fixed tick
Counterparty risk	You bear it. If they default, you litigate.	Novated to the clearing corporation, which becomes buyer to every seller and seller to every buyer
Margin	Negotiated or absent	Mandatory, computed by SPAN, collected upfront, topped up intraday
Daily settlement	None — P&L settles once, at maturity	Mark to market every day; cash moves daily
Liquidity / exit	You must renegotiate with the same counterparty	Sell to anyone in the order book
Who uses it	Corporates hedging currency, commodity producers, banks	Everyone, including individuals

> **The clearing corporation is the unsung hero**
> When you buy an index future, you are not relying on the stranger who sold it. The clearing corporation steps in between. It collects margin from both sides daily, marks positions to market, and maintains a settlement guarantee fund. This is why a retail participant can take the other side of a bank's position without doing credit analysis on the bank. It is also why margin rules are not negotiable — the guarantee only works if the margin is actually collected.

Indian equity futures: the specification

> **Verify every line of this table**
> Contract specifications in Indian derivatives — lot sizes, expiry days, which indices have weekly contracts — have changed several times since 2024 and may change again. The table below is the *shape* of a specification, with illustrative values **as at late 2026**. Get the live spec from the contract master at **nseindia.com** / **bseindia.com** before placing any order.

Specification	Index futures (illustrative)	Single-stock futures (illustrative)
Underlying	A benchmark or sectoral equity index	An individual listed share from the F&O eligible list
Lot size	Set by exchange; raised substantially by SEBI's 2024–25 framework so that one contract has a minimum notional in the several-lakh range	Set per stock so notional clears the minimum contract value threshold; revised periodically
Expiry cycle	Near month, next month, far month (3 serial months) for monthly contracts	Near, next, far month
Expiry day	A designated weekday of the expiry month — has been changed; verify	Same day as index monthly expiry
Tick size	₹0.05	₹0.05
Price quotation	Index points	Rupees per share
Settlement	**Cash** — difference settled in rupees against the settlement value	**Physical delivery** of shares — this is the one that catches beginners
Final settlement price	Typically a volume-weighted average of the underlying over the last period of the expiry day	Closing price of the underlying on expiry day
Daily settlement	Daily mark to market against the day's settlement price	Daily mark to market

The expiry calendar, and why it keeps moving

For years, Indian index options had weekly expiries on multiple indices across both exchanges, producing an expiry event almost every trading day of the week. That concentration of short-dated, cheap, high-gamma contracts is exactly where retail losses were heaviest. SEBI's October 2024 framework restricted weekly expiries to **one benchmark index per exchange**. Exchanges have since reshuffled which weekday their expiry falls on. The practical consequences:

- ☐ Never assume an expiry day from memory. Check the contract master.
- ☐ A strategy that depended on four weekly expiries a week no longer exists in that form.

- Monthly expiry is the last weekly expiry of the month for the relevant index — so monthly and weekly coincide that day, with correspondingly larger open interest and sharper moves.

Mark to market: the mechanic that surprises people

Futures do not settle once at expiry. They settle *every single day*. Cash physically moves in and out of your trading account at the end of each session. Worked example with a hypothetical index at 25,000 and lot size 75:

TEXT

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DAILY MARK-TO-MARKET ON ONE LONG INDEX FUTURES CONTRACT
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Lot size 75.  Entry: BUY 1 lot at 25,000 on Day 0.
Initial margin blocked (illustrative 12%): Rs 2,25,000

Day  Settlement  Change vs prev   MTM cash flow        Running
     price       settlement       (change x 75)        cash P&L
____ __________  _______________  ___________________  _________

 0    25,000      (entry price)    0                        0
 1    25,180      +180             +13,500 credited    +13,500
 2    24,990      -190             -14,250 debited        -750
 3    24,710      -280             -21,000 debited     -21,750
 4    24,600      -110              -8,250 debited     -30,000
 5    25,050      +450             +33,750 credited     +3,750

Observations that matter:
1. On Day 3 and Day 4 real money LEFT your account. If your free
   cash was insufficient, you get a margin call and may be squared
   off at the worst possible moment - at the Day 4 low, locking in
   -30,000 and missing the Day 5 recovery entirely.
2. Your "paper loss" is not paper. It is cash, daily.
3. The final position P&L (+50 points x 75 = +3,750) equals the sum
   of the daily flows. The path, however, decides whether you were
   still in the trade to collect it.

PATH RISK IS THE REAL RISK. Your directional call can be correct
and you can still be stopped out by the journey.
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"I'll just hold through the drawdown" is not available to you

In cash equity you can ignore a 20% fall indefinitely. In futures you cannot, because the loss is collected from your bank balance every evening and because a margin shortfall triggers a penalty and then an auto square-off. Leverage removes the option to be patient. That is its central cost.

Open interest, volume, and what they do and do not tell you

Metric	Definition	Honest interpretation
Volume	Contracts traded during the session	Liquidity. High volume means tight spreads and easy exit. That is almost all it tells you.
Open interest	Contracts still open (not squared off) at end of session	How much money is committed at that strike/contract. Rising OI = new positions being built; falling OI = positions being closed.
OI + price together	Four combinations: price up/OI up, price up/OI down, etc.	Commonly taught as "long build-up / short covering / short build-up / long unwinding". Treat with heavy scepticism — OI is net of all participants, you cannot see who is long, and a large OI build may be one leg of a hedged book, not a directional bet.
Max pain / PCR	Strike with maximum option-writer profit; put-call ratio of OI	Widely quoted, weakly evidenced. These are descriptive statistics, not forecasts. Nobody with a real edge is publishing it on a free dashboard.

> ⚠ **On "OI analysis"**
> An enormous amount of Indian retail content is built around reading open-interest tables to predict direction. The honest position: OI is a useful liquidity and positioning signal, and a very poor directional predictor. If OI reliably predicted direction, the firms with full order-flow visibility would have arbitraged that away long before the number reached a free website.

Futures pricing: cost of carry, basis and a hedge that actually works

A futures price is not a forecast. It is today's spot price adjusted for the cost of holding the underlying until expiry. Once you internalise that, half the mystique of the futures market evaporates and the remaining half is arithmetic.

The cost-of-carry model

TEXT

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FAIR VALUE OF A FUTURES CONTRACT
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F  = S x e^(r x t)        (continuous compounding form)
or approximately
F  = S x (1 + r x t) - D   (simple form, with dividends)

where
  F = fair futures price
  S = spot price of the underlying
  r = risk-free interest rate (annualised, decimal)
  t = time to expiry in years
  D = present value of dividends expected before expiry

INTUITION
─────────
If you want exposure to the index in one month you have two choices:
  (a) buy the basket today and hold it for a month
  (b) buy the futures contract

With (a) you tie up capital for a month (you forgo interest r)
but you RECEIVE any dividends (D) paid in that month.
With (b) you tie up only margin.

For the two routes to cost the same - i.e. for there to be no
arbitrage - the futures price must be higher than spot by the
forgone interest and lower by the dividends received.

WORKED EXAMPLE
──────────────
Spot index S                      25,000.00
Risk-free rate r                  6.50% p.a.  (illustrative)
Days to expiry                    30  →  t = 30/365 = 0.08219 yr
Expected dividend yield over
  the period, in index points     22.00

Interest component = 25,000 x 0.065 x 0.08219
                   = 25,000 x 0.0053425
                   = 133.56 points

Fair value F = 25,000 + 133.56 - 22.00
             = 25,111.56

So with the index at 25,000, a one-month future SHOULD trade
around 25,112. If it trades at 25,250, it is rich by 138 points.
If it trades at 25,040, it is cheap by 72 points.

Rates and dividend assumptions are illustrative.
As at late 2026 - verify current rates independently.
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Basis, contango and backwardation

TEXT

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BASIS = FUTURES PRICE − SPOT PRICE

Futures 25,112   Spot 25,000   →  Basis = +112  (POSITIVE)
Futures 24,940   Spot 25,000   →  Basis =  −60  (NEGATIVE)

POSITIVE BASIS (futures above spot) = CONTANGO
  Normal for equity index futures, because the cost of carry
  (interest) usually exceeds the dividend yield.

NEGATIVE BASIS (futures below spot) = BACKWARDATION
  In equity indices this typically appears when:
   - a large dividend is due before expiry (dividends subtract)
   - there is heavy short hedging pressure (institutions selling
     futures to hedge cash holdings, pushing futures below fair)
   - borrowing the stock to arbitrage is hard or expensive
  Backwardation is a positioning signal, not a crash forecast.

CONVERGENCE
___________
At expiry, the futures contract settles against the underlying.
Therefore basis must be ZERO at expiry, by construction.

   Days to expiry:  30    20    10     5     2     0
   Typical basis:  +112   +75   +38   +19    +8     0
                    \                             /
                     \__ decays roughly linearly __/
                        with time to expiry

This decay is DETERMINISTIC, unlike option time decay which
depends on volatility. It is why a long futures position held
to expiry in contango carries a small structural headwind, and
a short futures position carries a small structural tailwind.
Both are tiny compared to directional risk.
```



Basis convergence. The futures line and the spot line must meet at expiry because the contract settles against the underlying.

Calendar spreads

A calendar spread is simultaneously long one expiry and short another of the same underlying. You are trading the *basis between two expiries*, not direction. Because the two legs largely offset, exchanges historically granted a margin benefit on such spreads.

⚠ **The expiry-day calendar-spread margin benefit was removed**

SEBI's 2024–25 framework removed the calendar-spread margin benefit **on the day of expiry** for contracts expiring that day. The rationale: on expiry day the two legs stop behaving like a hedged pair — the expiring leg's delta collapses to 0 or 1 while the far leg does not — so treating them as offsetting understated real risk. If you ran spreads into expiry relying on reduced margin, that is gone. Verify the current treatment at nseindia.com and with your broker's margin calculator.

TEXT

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CALENDAR SPREAD - WHAT YOU ARE ACTUALLY TRADING
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Position: SELL near-month future @ 25,112
          BUY  far-month future  @ 25,240
          Spread = 128 points

You now have (nearly) zero directional exposure. If the index
rises 500 points, both legs move roughly 500 points and cancel.

What you profit from: the spread NARROWING (if you are short the
near and long the far, you want the far-near gap to shrink).

What actually moves the spread:
  - changes in the interest rate assumption
  - changes in expected dividends between the two expiries
  - institutional roll pressure in the last week before expiry
  - stock-borrow costs for single-stock calendars

Typical spread range: a few tens of index points.
Typical move you can capture: perhaps 10-30 points.
Transaction cost: 4 legs (2 in, 2 out) of brokerage + STT + fees.

RUN THE NUMBERS BEFORE YOU ROMANTICISE THIS. Four legs of cost
to chase a 20-point spread on a 75 lot = Rs 1,500 gross, against
perhaps Rs 300-600 of round-trip cost plus slippage on four fills.
Without the margin benefit it also ties up substantial capital.
```

A worked hedge: protecting a ₹10 lakh equity portfolio with index futures

TEXT

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SITUATION (hypothetical)
========================
You hold a diversified Indian equity portfolio.
Portfolio value                      Rs 10,00,000  (ten lakh)
Portfolio beta vs the index          1.10
  (i.e. historically, when the index moves 1%,
   this portfolio has moved about 1.10%)
Index spot                           25,000
Index futures (1 month)              25,112
Lot size                             75
Notional per futures lot = 25,112 x 75 = Rs 18,83,400

You want to hedge for 4 weeks across a results season.
You do NOT want to sell the portfolio (capital gains tax,
re-entry risk, you still like the holdings long term).

STEP 1 - COMPUTE THE HEDGE RATIO
--------------------------------

Beta-adjusted exposure = Portfolio value x Beta
                       = 10,00,000 x 1.10
                       = Rs 11,00,000

Number of lots to SHORT = Beta-adjusted exposure / Notional per lot
                        = 11,00,000 / 18,83,400
                        = 0.584 lots

PROBLEM: you cannot short 0.584 of a lot. You can short 0 or 1.
This is the "lot size granularity" problem and it is the single
biggest practical obstacle to retail hedging in India after the
2024-25 lot-size increases.

STEP 2 - DECIDE
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Option A: short 1 lot --> you are OVER-hedged
          (hedging Rs 18.83 lakh of exposure against Rs 11 lakh)
Option B: do not hedge with futures; use index options instead,
          or accept the risk, or reduce the position in cash.

Let us work Option A and see what over-hedging costs you.

STEP 3 - SCENARIO ANALYSIS (short 1 index futures lot at 25,112)
----------------------------------------------------------------

Index   Index   Portfolio move   Portfolio    Futures P&L      Net
level   move    (= 1.10 x idx)   P&L (Rs)     (25,112-F)x75    (Rs)
------  ------  ---------------  -----------  ---------------  ---------
22,500  -10.0%     -11.0%         -1,10,000   +1,95,900        +85,900
23,750   -5.0%      -5.5%           -55,000     +1,02,150      +47,150
24,500   -2.0%       -2.2%          -22,000       +45,900      +23,900
```

```
25,000    0.0%        0.0%                0        +8,400       +8,400
25,500   +2.0%       +2.2%          +22,000       -29,100       -7,100
26,250   +5.0%       +5.5%          +55,000       -85,350      -30,350
27,500  +10.0%      +11.0%        +1,10,000     -1,79,100      -69,100

READ THE TABLE HONESTLY
_______________________

1. The hedge WORKS on the downside - a 10% crash costs you
   Rs 1.1 lakh on the portfolio but the short future makes
   Rs 1.96 lakh. You end up ahead.
2. But you are OVER-HEDGED. A 10% rally costs you Rs 69,100 net.
   You have converted a long portfolio into a net-short position.
3. The +8,400 at an unchanged index is the basis you captured by
   shorting the future above spot (112 points x 75). That is the
   one small structural tailwind of a short hedge in contango.
4. A hedge should NEUTRALISE risk, not reverse it. Over-hedging
   by 71% is not a hedge; it is a new directional bet.

MARGIN COST OF THE HEDGE
________________________

Short 1 index futures lot needs roughly
  SPAN + exposure margin, illustrative 12% of notional
  = 18,83,400 x 0.12 = Rs 2,26,008 blocked

Plus you must hold free cash for adverse daily MTM. If the index
rallies 5%, you pay out Rs 85,350 in daily MTM over those sessions
even though your portfolio is up - you need that cash available.

So hedging Rs 10 lakh of equity requires roughly Rs 2.26 lakh of
margin PLUS a cash buffer of perhaps Rs 1 lakh for MTM. That is
over 30% of the portfolio value sitting idle.

Margin figures illustrative. As at late 2026 - verify on the
exchange margin calculator at nseindia.com.
```

The honest conclusion on futures hedging for small portfolios

With post-2024 lot sizes, one index futures contract has a notional of roughly ₹15–20 lakh. If your portfolio is under about ₹15 lakh you cannot hedge it with index futures without badly over- or under-hedging. Index **options** are the finer instrument for small portfolios — a put gives you asymmetric protection and a known maximum cost, though you pay for it. The section on legitimate use cases works that example properly.

✦ **A hedge that makes money is usually a hedge you did not need**
If your hedge is profitable, your portfolio is down. The hedge's job is to reduce the variance of the combined outcome, not to produce P&L. If you find yourself pleased that the hedge is "winning", you have stopped hedging and started speculating.

Options fundamentals

An option is a contract that gives the buyer the **right but not the obligation** to buy (call) or sell (put) the underlying at a fixed price (the strike) on or before a fixed date (expiry), in exchange for a payment (the premium) made to the seller, who takes on the matching **obligation**. Every option concept in this guide is a consequence of that one sentence.

The four basic positions

Position	You pay or receive	Your right / obligation	You want the underlying to	Max profit	M
Long call (buy call)	Pay premium	Right to buy at K	Rise, a lot, soon	Theoretically unlimited	Pr
Short call (write call)	Receive premium	Obligation to sell at K if assigned	Fall, stay flat, or rise slowly	Premium received	**Tl ur**
Long put (buy put)	Pay premium	Right to sell at K	Fall, a lot, soon	(K − premium) × lot, if underlying → 0	Pr
Short put (write put)	Receive premium	Obligation to buy at K if assigned	Rise, stay flat, or fall slowly	Premium received	(K lo

Read the "max loss" column again

Two of the four basic positions have effectively unlimited or near-unlimited loss. Those two are the ones that "win" most of the time. That combination — high win rate, unbounded loss — is precisely the structure that destroys accounts, because a long run of small wins builds confidence and position size right up to the moment of the single catastrophic loss. This is quantified later with a worked blow-up scenario.

Premium: what you are quoting

TEXT

```
READING AN OPTION QUOTE
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Suppose the option chain shows, with the index at 25,000:

  Strike   Type   Premium (points)
  25,000   CE     218.50
  25,000   PE     196.00
  25,300   CE      96.25
  24,700   PE      82.75

"CE" = Call European, "PE" = Put European.
Premium is quoted in INDEX POINTS, not rupees.

RUPEE COST = premium x lot size
  Buying 1 lot of the 25,000 CE at 218.50 with lot size 75:
    218.50 x 75 = Rs 16,387.50  (plus charges)

  Buying 1 lot of the 25,300 CE at 96.25:
    96.25 x 75 = Rs 7,218.75   (plus charges)

NOTIONAL CONTROLLED by that Rs 16,387.50:
    25,000 x 75 = Rs 18,75,000

So Rs 16,388 of premium controls Rs 18.75 lakh of notional for
the life of the contract. That is roughly 114x notional leverage.
It is also why the premium can go to zero: you are paying for a
contingent claim that expires worthless unless the index moves
far enough, fast enough.

Premiums illustrative. As at late 2026 - verify live quotes.
```

European versus American

	European	American
When can the buyer exercise?	Only at expiry	Any time up to expiry
Early assignment risk for the writer?	No	Yes
Indian equity index options	**European**	—
Indian single-stock options	**European** (converted from American years ago)	—

> **Why "European" simplifies your life**
> Because Indian listed equity options are European-style, you cannot be assigned early. A short option position cannot be exercised against you on a random Tuesday — only at expiry. This removes one whole category of risk that American-style markets have. It does **not** remove assignment risk at expiry, and for single-stock options that assignment means **physical delivery of shares**, which is covered in detail later and is the most common way a beginner gets a nasty surprise. Verify current style and settlement rules at nseindia.com.

> **You rarely "exercise" anyway**
> In practice almost nobody exercises an option manually. You close the position by selling the option back into the market before expiry, capturing whatever premium remains. Exercise happens automatically at expiry for in-the-money contracts. Trading out is almost always better than exercising, because exercising throws away any remaining time value.

Moneyness

TEXT

```
MONEYNESS — index spot at 25,000
═══════════════════════════════════

                CALLS                          PUTS
Strike   Relationship    State     |   Relationship    State
──────   ──────────────  ────────  |   ──────────────  ────────
24,400   K far below S   deep ITM  |   K far below S   deep OTM
24,700   K below S       ITM       |   K below S       OTM
25,000   K = S           ATM       |   K = S           ATM
25,300   K above S       OTM       |   K above S       ITM
25,600   K far above S   deep OTM  |   K far above S   deep ITM

ITM  = In The Money   — has intrinsic value right now
ATM  = At The Money   — strike is at (or nearest) the spot
OTM  = Out of The Money — zero intrinsic value; pure time value

RULE OF THUMB
  A call is ITM when  Spot > Strike
  A put  is ITM when  Spot < Strike
  (They are mirror images. There is nothing more to it.)
```

Intrinsic value and time value — worked

TEXT

DECOMPOSING THE PREMIUM

PREMIUM = INTRINSIC VALUE + TIME VALUE (a.k.a. extrinsic value)

INTRINSIC VALUE

- Call: max(Spot − Strike, 0)
- Put : max(Strike − Spot, 0)
- It can never be negative. It is the value you would get if the contract expired RIGHT NOW.

TIME VALUE = Premium − Intrinsic value

- Everything you are paying ABOVE the immediate exercise value.
- It compensates the writer for the risk that the underlying moves further in your favour before expiry.
- At expiry, time value is ZERO, always, for every option.

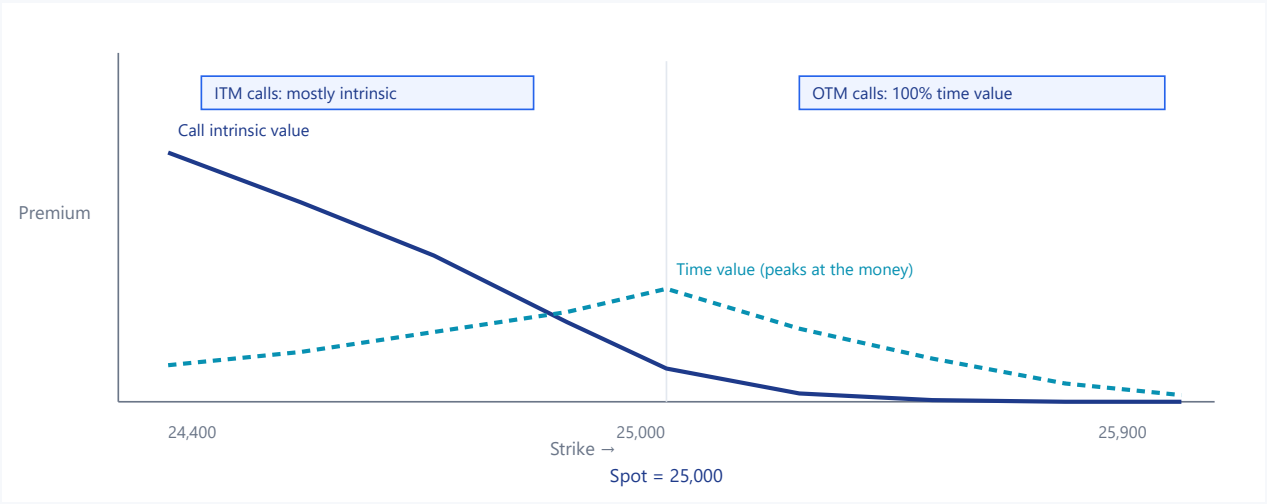
WORKED − index spot 25,000, 18 days to expiry

Contract	Premium	Intrinsic	Time value
24,700 CE	382.00	25,000 − 24,700 = 300.00	82.00
25,000 CE	218.50	max(0,0) = 0.00	218.50
25,300 CE	96.25	max(0,−300) = 0.00	96.25
25,600 CE	34.50	max(0,−600) = 0.00	34.50
24,700 PE	82.75	max(0,−300) = 0.00	82.75
25,000 PE	196.00	max(0,0) = 0.00	196.00
25,300 PE	383.60	25,300 − 25,000 = 300.00	83.60
25,600 PE	629.00	25,600 − 25,000 = 600.00	29.00

THREE THINGS TO NOTICE

1. The ATM options have the MOST time value (218.50 and 196.00). Time value peaks at the money and falls away in both directions.
2. Deep ITM options have very little time value (29.00 on the 25,600 PE) − they behave almost like the underlying itself.
3. The OTM options are 100% time value. Buy a 25,600 CE for 34.50 and if the index closes at 25,000 on expiry, you lose 100%. Not 20%. One hundred percent.

Premiums illustrative. As at late 2026.



Premium splits into intrinsic value (a hard floor set by spot versus strike) and time value (maximal at the money, decaying to zero at expiry).

Call and put parity — the relationship that must hold

TEXT

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PUT-CALL PARITY (European options, same strike and expiry)
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  C − P  =  S − K x e^(−r x t)

  C = call premium, P = put premium
  S = spot, K = strike, r = rate, t = time in years

CHECK IT with our numbers (spot 25,000, strike 25,000,
18 days, r = 6.5%):

  t = 18/365 = 0.049315
  K x e^(−r x t) = 25,000 x e^(−0.065 x 0.049315)
                 = 25,000 x e^(−0.003205)
                 = 25,000 x 0.996800
                 = 24,920.00

  Right-hand side  = 25,000 − 24,920 = 80.00
  Left-hand side   = C − P = 218.50 − 196.00 = 22.50

The two do not match exactly (80.00 vs 22.50) because our
illustrative premiums ignore the dividend the index will pay
before expiry. Adding an expected dividend of about 57 index
points to the put side closes the gap.

WHY YOU CARE
────────────
1. Parity is why a synthetic position exists: long call + short
   put at the same strike = long futures. Any time you think you
   have invented a clever structure, check whether parity says it
   is just a future in disguise with extra transaction costs.
2. Large persistent parity violations do not exist in liquid
   contracts. If your screen shows one, it is a stale quote, a
   dividend you forgot, or an illiquid strike you cannot actually
   trade at the displayed price.
3. It means calls and puts are not independent instruments. If you
   have a view on one, you implicitly have a view on the other.
```

Payoff diagrams and break-even arithmetic

A payoff diagram shows profit or loss on the vertical axis against the underlying price at expiry on the horizontal axis. It ignores time value entirely — it is the picture *at expiry*. That is a critical limitation and the source of a lot of beginner confusion, which this section addresses directly.

⚠ **The diagram is the end of the film, not the film**

Payoff diagrams describe the position's value **at expiry only**. Before expiry, the position is worth something different because of remaining time value and implied volatility. A long call can be deeply underwater on the diagram and still be worth money today. A short straddle can show a comfortable profit zone and still be bleeding badly mid-cycle because volatility rose. Never size a position using only the expiry diagram.

Long call



Long call at expiry. Loss is capped at the premium; profit is open-ended above the break-even.

TEXT

LONG CALL – FULL PAYOFF TABLE

Position : BUY 1 lot 25,000 CE @ 218.50, lot size 75
Cost   : 218.50 x 75 = Rs 16,387.50 (ignore charges for now)

Payoff per unit at expiry = max(Spot – 25,000, 0) – 218.50
Break-even = Strike + Premium = 25,000 + 218.50 = 25,218.50

Spot at expiry	Intrinsic per unit	Payoff per unit	P&L in Rs (x 75)	Return on premium	
23,500	0.00	-218.50	-16,387.50	-100.0%	
24,000	0.00	-218.50	-16,387.50	-100.0%	
24,500	0.00	-218.50	-16,387.50	-100.0%	
24,900	0.00	-218.50	-16,387.50	-100.0%	
25,000	0.00	-218.50	-16,387.50	-100.0%	
25,100	100.00	-118.50	-8,887.50	-54.2%	
25,218.50	218.50	0.00	0.00	0.0%	← BE
25,300	300.00	81.50	6,112.50	+37.3%	
25,500	500.00	281.50	21,112.50	+128.8%	
26,000	1,000.00	781.50	58,612.50	+357.7%	
26,500	1,500.00	1,281.50	96,112.50	+586.3%	

THE PART NOBODY PUTS ON THE BROCHURE

The index must rise 0.874% (218.50 / 25,000) JUST TO BREAK EVEN.
If the index finishes anywhere from 0 to 25,000 – which covers the entire down side and flat – you lose 100% of the premium.
If it finishes between 25,000 and 25,218.50 you still lose money even though your directional call was CORRECT.

Probability-weighted, with 18 days left, the chance of finishing above 25,218.50 is somewhere near 40% for an at-the-money-ish option on a typical index. That is the whole trade: roughly a 60% chance of losing some or all of the premium.

Short call (naked)



Short naked call. The mirror image of the long call: a small capped gain against an unbounded loss. Note that the loss line does not stop at the edge of the chart.

TEXT

SHORT NAKED CALL – FULL PAYOFF TABLE
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Position : SELL 1 lot 25,000 CE @ 218.50, lot size 75
Received : Rs 16,387.50
Margin blocked (illustrative): Rs 1,90,000 – 2,50,000
  (far more than the premium received – verify on the exchange
   margin calculator; as at late 2026)

Payoff per unit = 218.50 – max(Spot – 25,000, 0)
Break-even = 25,218.50

Spot at expiry	Payoff per unit	P&L in Rs (x 75)	Return on margin (assume Rs 2,20,000)	
23,500	+218.50	+16,387.50	+7.4%	
24,500	+218.50	+16,387.50	+7.4%	
25,000	+218.50	+16,387.50	+7.4%	
25,218.50	0.00	0.00	0.0%	← BE
25,500	-281.50	-21,112.50	-9.6%	
26,000	-781.50	-58,612.50	-26.6%	
26,500	-1,281.50	-96,112.50	-43.7%	
27,000	-1,781.50	-1,33,612.50	-60.7%	
28,000	-2,781.50	-2,08,612.50	-94.8%	
29,000	-3,781.50	-2,83,612.50	-128.9%	← you now owe money beyond the margin

THE ARITHMETIC THAT SHOULD STOP YOU
──────────────────────────────────

Best case ever: +Rs 16,387
A 16% index rally: -Rs 2,83,612
Ratio: you risk 17x your maximum gain

A 16% single-contract-cycle index move is rare. It is not
impossible. Indian indices have moved more than that inside a
month on multiple occasions in the last two decades.

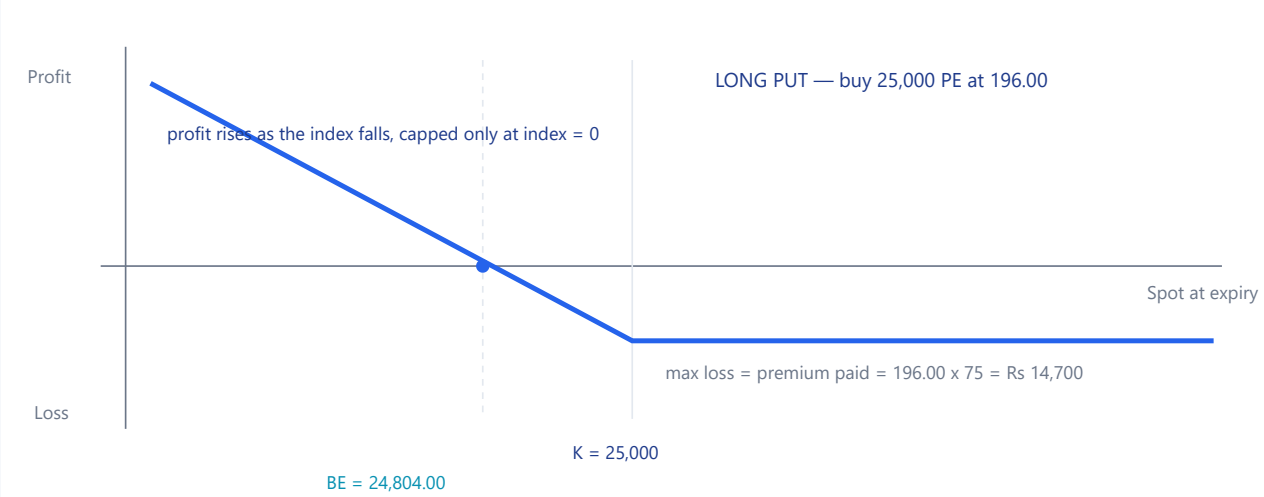
To make Rs 16,387 seventeen times in a row without one loss of
that size, you need the move NOT to happen seventeen cycles
running. Then it happens once, and you are back to zero – minus
all the transaction costs of the seventeen winning cycles.

THIS IS THE ENTIRE "SELL OPTIONS FOR MONTHLY INCOME" PITCH,
written out with the numbers attached.

Naked short calls

A short call with no offsetting long position has unbounded loss. Many Indian brokers will not permit it without substantial margin, and the exchange margin requirement is deliberately punitive. Treat the margin requirement as the exchange telling you what it thinks of the risk. For a retail participant with a single account and no risk desk, there is no defensible reason to hold a naked short call.

Long put



Long put at expiry. The insurance contract: a known, limited cost against a large but finite payoff if the market falls.

TEXT

LONG PUT – FULL PAYOFF TABLE

Position : BUY 1 lot 25,000 PE @ 196.00, lot size 75
Cost   : 196.00 x 75 = Rs 14,700.00

Payoff per unit = max(25,000 – Spot, 0) – 196.00
Break-even = Strike – Premium = 25,000 – 196.00 = 24,804.00

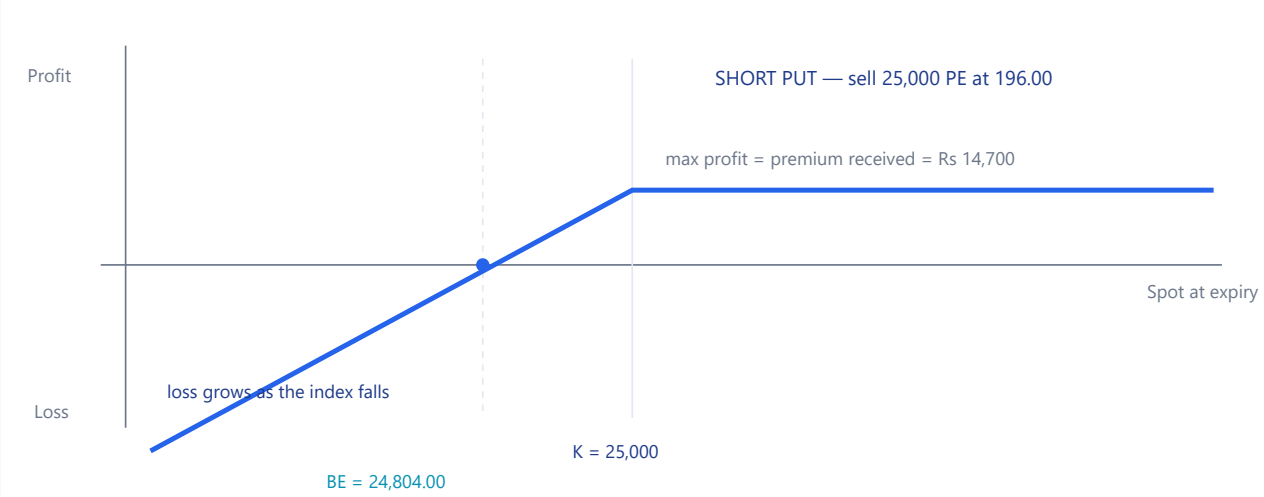
Spot at expiry	Intrinsic per unit	Payoff per unit	P&L in Rs (x 75)	Return on premium	
26,500	0.00	-196.00	-14,700.00	-100.0%	
25,500	0.00	-196.00	-14,700.00	-100.0%	
25,000	0.00	-196.00	-14,700.00	-100.0%	
24,900	100.00	-96.00	-7,200.00	-49.0%	
24,804.00	196.00	0.00	0.00	0.0%	← BE
24,700	300.00	104.00	7,800.00	+53.1%	
24,500	500.00	304.00	22,800.00	+155.1%	
24,000	1,000.00	804.00	60,300.00	+410.2%	
23,000	2,000.00	1,804.00	1,35,300.00	+920.4%	
22,500	2,500.00	2,304.00	1,72,800.00	+1,175.5%	

Maximum theoretical profit (index to zero):
  (25,000 – 196.00) x 75 = Rs 18,60,300 – which will not happen.
  Puts are "limited" profit only in the pedantic sense.

THE HONEST VIEW

This is the one long-option position with a genuine, defensible retail use: buying downside insurance on a portfolio you own. You pay Rs 14,700 for the right to sell at 25,000 for 18 days. Like any insurance, you EXPECT to lose the premium most of the time. That is not a failure of the trade; that is what insurance costs. Buying puts as a DIRECTIONAL BET is a different activity and has the same ~60% loss probability as buying calls.

Short put (cash-secured or naked)



Short put at expiry. Looks benign because the left tail is bounded by the index reaching zero — but the bound is so far away that it is unbounded for practical purposes.

TEXT

```
SHORT PUT – FULL PAYOFF TABLE
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Position : SELL 1 lot 25,000 PE @ 196.00,  lot size 75
Received : Rs 14,700.00
Margin (illustrative): Rs 1,85,000 – 2,40,000 blocked

Spot at   Payoff      P&L in Rs      Return on margin
expiry    per unit    (x 75)         (assume Rs 2,10,000)
────────  ──────────  ─────────────  ────────────────────
26,000      +196.00     +14,700.00         +7.0%
25,500      +196.00     +14,700.00         +7.0%
25,000      +196.00     +14,700.00         +7.0%
24,804.00      0.00           0.00          0.0%  ← BE
24,500      -304.00     -22,800.00        -10.9%
24,000      -804.00     -60,300.00        -28.7%
23,500    -1,304.00     -97,800.00        -46.6%
23,000    -1,804.00  -1,35,300.00        -64.4%
22,000    -2,804.00  -2,10,300.00       -100.1%  ← margin gone
20,000    -4,804.00  -3,60,300.00       -171.6%

A 12% index fall in one contract cycle wipes the margin.
Indian indices have fallen more than 12% in a month.
In March 2020 benchmark indices fell roughly 25% in weeks.
A 25% fall here = (25,000 – 18,750 – 196) x 75 = -Rs 4.54 lakh
on a position whose maximum possible gain was Rs 14,700.
```

⚠ **"Cash-secured put" is a meaningfully different trade**

If you set aside the full ₹18,75,000 (25,000 × 75) in cash and genuinely intend to take the exposure at 25,000, selling the put is a legitimate way to get paid for a limit order you were going to place anyway. Your effective entry becomes 24,804. If instead you post ₹2.1 lakh of margin and have no intention of taking delivery, it is a leveraged short-volatility bet with a 12.7x loss-to-gain ratio wearing a respectable name. The structure is identical; the collateral and the intent are what differ.

Summary of the four legs

Leg	Direction view	Volatility view	Time decay	Break-even	Ri
Long call	Bullish	Long vol (wants IV up)	Against you	K + premium	Ca op
Short call	Bearish/neutral	Short vol	For you	K + premium	Ca op
Long put	Bearish	Long vol	Against you	K − premium	Ca la
Short put	Bullish/neutral	Short vol	For you	K − premium	Ca la

✦ **Notice the pattern**

Every option position with time decay *working for you* also has the open-ended loss. Every position with capped loss is paying time decay. There is no leg where both are favourable, because if there were, everyone would put it on and the premium would adjust until it was not. Multi-leg strategies are attempts to trade off these four properties against each other — not to escape them.

What drives an option premium: volatility, IV and the India VIX

Six inputs determine a European option's theoretical price. Five of them you can look up. The sixth — volatility — you cannot, and that is where essentially all the money is made and lost in options.

The six inputs

Input	Symbol	Effect on a CALL	Effect on a PUT	Observable?
Spot price	S	Up → call up	Up → put down	Yes
Strike price	K	Higher K → cheaper call	Higher K → dearer put	Yes (you choose it)
Time to expiry	t	More time → dearer	More time → dearer	Yes
Risk-free rate	r	Higher r → slightly dearer	Higher r → slightly cheaper	Yes
Dividends	D	More dividend → cheaper	More dividend → dearer	Mostly (estimated)
Volatility	σ	**Higher → dearer**	**Higher → dearer**	**NO — it is a forecast**

> **Volatility raises both calls and puts**
> This is counter-intuitive until you see why. Volatility measures the *spread* of possible outcomes, not their direction. A wider spread of outcomes increases the chance of finishing far in the money, and an option's downside is already capped at zero — the buyer cannot lose more by the underlying going further the wrong way. More uncertainty is therefore unambiguously good for an option *buyer* and bad for a *seller*, on both sides of the chain.

Black–Scholes, conceptually

The Black–Scholes–Merton model prices a European option by constructing a hypothetical portfolio of the underlying and cash that replicates the option's payoff, then arguing that the option must cost what that replicating portfolio costs. You do not need the derivation. You need the shape of the answer and the places it is wrong.

TEXT

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BLACK-SCHOLES: WHAT IT SAYS IN WORDS
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Call price = (probability-weighted value of receiving the stock
              if it finishes above the strike)
           - (discounted strike, weighted by the probability of
              finishing above the strike)

Formally:   C = S x N(d1) - K x e^(-rt) x N(d2)

   N(.)  = cumulative normal distribution function
   N(d2) = (risk-neutral) probability the option expires ITM
   N(d1) = the option's DELTA (see the Greeks section)

   d1 = [ ln(S/K) + (r + sigma^2 / 2) x t ] / (sigma x sqrt(t))
   d2 = d1 - sigma x sqrt(t)

YOU WILL NEVER COMPUTE THIS BY HAND. Your broker's platform and
every option calculator does it. What matters is the ASSUMPTIONS,
because each one is false in the Indian market:

ASSUMPTION                      REALITY
_______________________________  _____________________________________

Returns are lognormal, no jumps Indices gap on budgets, results,
                                policy news, global shocks.
                                Fat tails are the norm.
Volatility is constant          Volatility clusters: calm periods
                                and violent periods alternate.
No transaction costs            STT, brokerage, GST, stamp duty,
                                spread - all material for retail.
Continuous hedging is possible  You cannot rehedge continuously,
                                and each rehedge costs money.
You can borrow/lend at r        You cannot.

CONSEQUENCE: Black-Scholes is a QUOTING CONVENTION, not a truth.
The market does not use it to find the price. The market finds
the price, then inverts the formula to express that price as a
volatility number. That number is IMPLIED VOLATILITY.
```

Implied volatility: the only number that is really traded

TEXT

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IMPLIED VOLATILITY (IV)
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Take the market premium. Hold S, K, t, r, D fixed. Ask:
"what value of sigma makes Black-Scholes output this premium?"
That sigma is the IMPLIED volatility.

IV is NOT a prediction you should trust. It is the market's
consensus price for uncertainty, expressed in annualised
percentage terms.

TRANSLATING IV INTO AN EXPECTED MOVE
────────────────────────────────────

Expected 1-standard-deviation move over period t:

   Move = Spot x IV x sqrt(t)

Worked, index at 25,000, IV = 14%, 18 days to expiry:

   t = 18/365 = 0.049315
   sqrt(t) = 0.22207
   Move = 25,000 x 0.14 x 0.22207
        = 777.2 index points

INTERPRETATION: with IV at 14%, the options market is pricing
roughly a 68% chance that the index finishes within
  25,000 +/- 777  =  24,223 to 25,777  in 18 days,
and roughly a 95% chance within +/- 1,554 (two SDs).

SANITY CHECK ON YOUR OWN TRADE
──────────────────────────────

Before buying a 25,600 CE (600 points OTM) with 18 days left:
  600 / 777 = 0.77 standard deviations.
  Probability of finishing above 0.77 SD is roughly 22%.
  Probability of finishing above the BREAK-EVEN (25,600 + 34.50
  = 25,634.50, i.e. 0.82 SD) is roughly 21%.

So you have about a 1-in-5 chance of making anything at all.
The premium is not "cheap at Rs 34.50". It is priced for a
1-in-5 event. The market knows what it is selling you.

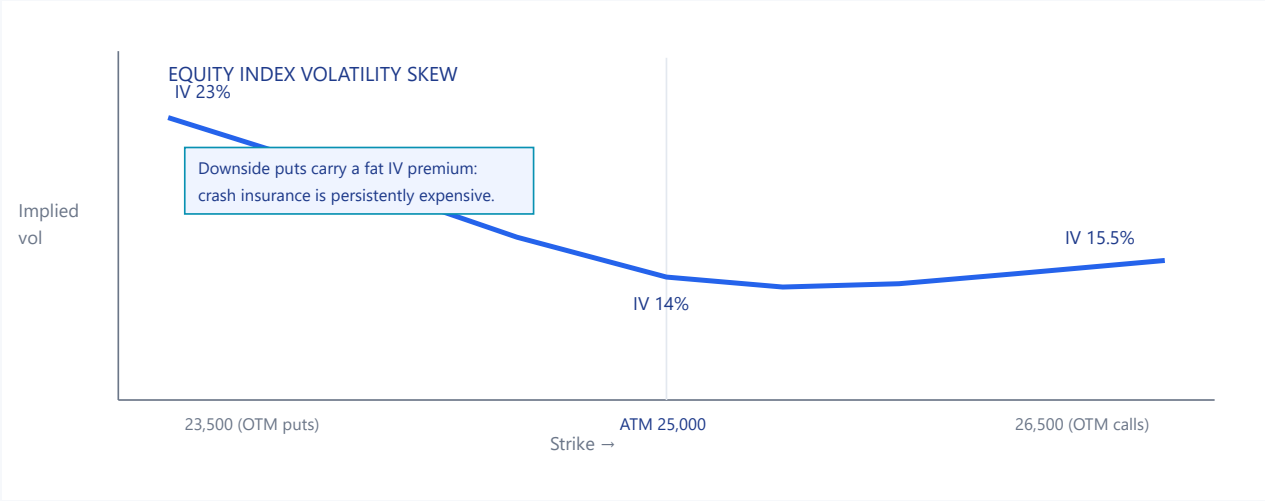
IV figures illustrative. As at late 2026 - verify live IV on the
exchange option chain at nseindia.com.
```

The single most useful IV habit

Before every option trade, compute the expected 1-SD move and compare it to the distance from spot to your break-even. If your break-even is more than one standard deviation away, you are buying a low-probability lottery ticket and should say so out loud. If it is less than half a standard deviation away, you are paying heavily for a high-probability outcome and your reward-to-risk is correspondingly poor. There is no free strike.

The volatility smile and skew

If Black–Scholes were correct, every strike on the same expiry would have the same IV. They do not. Plotting IV against strike produces a curve — a "smile" or, in equity indices, a downward-sloping "skew".



A typical equity-index skew. Out-of-the-money puts trade at materially higher implied volatility than out-of-the-money calls because markets fall faster than they rise and demand for downside protection is structural.

Observation	Why it happens	What it means for you
OTM puts have the highest IV	Equity markets crash down, drift up. Institutions buy downside protection persistently.	Downside protection is expensive. Budget for that when hedging. Also: selling those "cheap-looking, far-away" puts is selling the fattest-tailed risk in the chain.
ATM has the lowest IV	Most liquid, most competitively priced, least tail content.	ATM options are usually the fairest-priced part of the chain.
IV rises before known events	Results, policy announcements, budgets, elections create a known uncertainty spike.	You pay up for optionality across an event. If the event resolves without a big move, IV collapses and you lose even if direction was right.
Short-dated IV is more volatile than long-dated	Less time to average out; gamma is concentrated.	Weekly options are where IV swings hardest. That is attractive to professionals and lethal to beginners.

India VIX

India VIX is a volatility index computed by NSE from the order book of near-term index option contracts. It expresses, as an annualised percentage, the volatility the options market is pricing for roughly the next 30 days.

TEXT

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USING INDIA VIX
═══════════════
VIX is quoted annualised. Convert to the horizon you care about:

  Daily expected move   = VIX / sqrt(252)   [252 trading days]
  Monthly expected move = VIX / sqrt(12)
  N-day expected move   = VIX x sqrt(N/365)   [calendar basis]

WORKED, VIX = 14:
  Daily   = 14 / 15.87 = 0.88%   → about 220 index pts at 25,000
  Monthly = 14 / 3.464 = 4.04%   → about 1,010 index pts
  18-day  = 14 x sqrt(18/365) = 14 x 0.2221 = 3.11% → 777 pts

WORKED, VIX = 24 (a stressed market):
  Daily   = 1.51%   → about 378 index pts
  18-day  = 5.33%   → about 1,333 index pts

WHAT VIX DOES AND DOES NOT TELL YOU
___________________________________
DOES:  how expensive options are right now, in relative terms.
       how big a daily move the market is braced for.
DOES NOT: which direction. VIX is directionless by construction.
DOES NOT: when volatility will change. "VIX is low so a crash is
          coming" is not a tradeable statement; VIX can stay low
          for a year.

HISTORICAL BEHAVIOUR (shape, not a forecast)
  Calm regime        : VIX roughly 10-14
  Normal             : 14-18
  Nervous            : 18-25
  Crisis             : 30-80+ (March 2020 saw extraordinary levels)

VIX is mean-reverting over long horizons and explosively
non-mean-reverting over short ones. Selling volatility because
"VIX is high so it must come down" is the classic blow-up trade.

As at late 2026 - verify current India VIX methodology and levels
at nseindia.com.
```

IV crush: the mechanism that kills expiry-day buyers

TEXT

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WORKED EXAMPLE: RIGHT ON DIRECTION, WRONG ON P&L
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SETUP (hypothetical)
  Index spot                        25,000
  A major policy announcement is due in 2 days.
  IV on the near-expiry chain has been bid up to 22%
  (versus a typical 14% in calm conditions).
  You buy 1 lot of the 25,200 CE.
  Days to expiry                    6
  Premium at IV 22%                 Rs 148.00 per unit
  Cost = 148 x 75 = Rs 11,100

THE EVENT HAPPENS. The index rises 180 points to 25,180.
You were RIGHT about the direction.

BUT: the uncertainty is now resolved. IV collapses from 22% to 13%.
Time to expiry has also fallen from 6 days to 4.

NEW PREMIUM, 25,200 CE, spot 25,180, 4 days left, IV 13%:
  Roughly Rs 96.00 per unit

YOUR P&L
  Entry  148.00
  Exit    96.00
  Loss   -52.00 per unit  x 75  =  -Rs 3,900   (-35%)

DECOMPOSING WHAT HAPPENED
  Gain from spot moving +180 pts (delta effect)   +Rs  71 /unit
  Loss from IV falling 22% → 13% (vega effect)    -Rs 102 /unit
  Loss from 2 days passing (theta effect)         -Rs  21 /unit
  ──────────────────────────────────────────────────────────
  Net                                             -Rs  52 /unit

YOU WERE RIGHT ABOUT DIRECTION AND LOST 35%.

This is routine. It is not bad luck. The market priced the event
INTO the premium before the event. You bought the uncertainty at
its most expensive moment and sold it at its cheapest.

THE GENERAL RULE
  Buying options immediately before a known, scheduled event is
  buying high IV. You need the move to exceed what is ALREADY
  priced in - not merely to go the right way.
```

✕ **Expiry-day option buying**

On expiry day, the remaining premium of an out-of-the-money option is almost entirely time value that must decay to exactly zero within hours. Theta is at its maximum and gamma makes the position violently sensitive. The product is cheap in rupees (₹5, ₹12, ₹30 per unit) which makes it feel low-risk. The probability of any given OTM expiry-day option finishing in the money is low — often in the single digits to low teens — and the loss when it does not is 100%, not 20%. This is where SEBI's data shows the heaviest concentration of small-account losses. The cheapness is the trap: it encourages repeated small bets with a deeply negative expected value, which compounds relentlessly.

The Greeks

The Greeks are partial derivatives: each one answers "if exactly one input moves by one unit and everything else is held still, how much does my option's value change?" They are the vocabulary of risk for anyone holding options, and they are the difference between knowing what you own and guessing.

Greek	Measures sensitivity to	Units	Long option sign	Short option sign
Delta (Δ)	Underlying price	Premium change per 1 point of underlying	Call +, Put −	Call −, Put +
Gamma (Γ)	Delta itself (rate of change of delta)	Delta change per 1 point of underlying	Always +	Always −
Theta (Θ)	Passage of time	Premium change per calendar day	Always −	Always +
Vega (ν)	Implied volatility	Premium change per 1 percentage point of IV	Always +	Always −
Rho (ρ)	Interest rate	Premium change per 1 percentage point of rate	Call +, Put −	Call −, Put +

> **"Vega" is not a Greek letter**
> Delta, gamma, theta and rho are Greek. Vega is not — it is a trading-desk coinage. Nobody minds. Mention it only to be the most pedantic person in the room.

Delta

TEXT

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DELTA - THREE SIMULTANEOUS MEANINGS
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1. RATE OF CHANGE
   "If the index moves 1 point, my option premium moves by
    DELTA points."

2. EQUIVALENT UNDERLYING POSITION ("delta equivalent")
   A 0.50 delta call on a 75 lot behaves like owning
   0.50 x 75 = 37.5 units of the index.

3. ROUGH PROBABILITY OF FINISHING IN THE MONEY
   A 0.30 delta call has roughly a 30% chance of expiring ITM.
   (This is an approximation of N(d2), not N(d1), but close
    enough for position sizing intuition.)

RANGES
   Long call  delta: 0 to +1
   Short call delta: 0 to -1
   Long put   delta: 0 to -1
   Short put  delta: 0 to +1
   Long 1 futures lot: delta = +1 per unit

WORKED - index at 25,000, 18 days to expiry, IV 14%
──────────────────────────────────────────────────

Strike  Type  Premium   Delta   Delta-equiv units (x75 lot)
──────  ────  ────────  ──────  ──────────────────────────
24,400  CE      668.00   0.86            +64.5
24,700  CE      382.00   0.72            +54.0
25,000  CE      218.50   0.52            +39.0
25,300  CE       96.25   0.30            +22.5
25,600  CE       34.50   0.13             +9.8
25,900  CE       10.20   0.05             +3.8

24,400  PE       46.00  -0.14            -10.5
24,700  PE       82.75  -0.28            -21.0
25,000  PE      196.00  -0.48            -36.0
25,300  PE      383.60  -0.70            -52.5
25,600  PE      629.00  -0.87            -65.3

WORKED P&L FROM DELTA ALONE
──────────────────────────

Hold 1 lot of 25,300 CE (delta 0.30, premium 96.25).
Index rises 60 points to 25,060.

Approximate new premium = 96.25 + (0.30 x 60) = 96.25 + 18.00
                        = 114.25
P&L = 18.00 x 75 = +Rs 1,350
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Now the SAME 60-point move on 1 lot of 25,000 CE (delta 0.52):
P&L = 0.52 x 60 x 75 = +Rs 2,340

And on 1 lot of 24,400 CE (delta 0.86):
P&L = 0.86 x 60 x 75 = +Rs 3,870

SAME VIEW. SAME MOVE. THREE DIFFERENT OUTCOMES.
Delta is how much of the move you actually own. "Cheap" OTM
options are cheap because you are buying very little of the move.

Greeks illustrative. Use your broker's option calculator for live
values. As at late 2026.
```

✦ **Delta is the honest answer to "how many lots?"**
If you would be comfortable holding 40 units of index exposure, then one 0.52-delta call (39 delta-equivalent units) is roughly the right size and three 0.13-delta calls (29 units) is roughly a third smaller than it feels. Beginners size by rupee premium ("I only spent ₹7,000") instead of by delta, which is why positions are routinely 3–5x larger in risk terms than intended.

Gamma

TEXT

GAMMA — THE RATE OF CHANGE OF DELTA

Gamma tells you how fast your exposure changes as the market moves.
High gamma = your position's character changes rapidly.

WHERE GAMMA LIVES

- Highest: at the money, close to expiry
- Lowest : deep ITM, deep OTM, or far from expiry

Index 25,000, IV 14%. Gamma per 1-point move:

Strike	Type	18 days	7 days	2 days	Expiry day (3 hrs)
24,400	CE	0.00018	0.00009	0.00001	0.00000
25,000	CE	0.00056	0.00092	0.00171	0.00480
25,600	CE	0.00021	0.00014	0.00002	0.00000

(Index-point gammas are small numbers; multiply by the move
 and the lot to get something you can feel.)

WORKED — WHY GAMMA MATTERS ON EXPIRY DAY

Position: SHORT 1 lot of the ATM 25,000 CE on expiry morning.
Premium Rs 42.00. Delta −0.50. Gamma 0.0048.

Index moves up 100 points to 25,100.
  New delta = −0.50 − (0.0048 x 100) = −0.50 − 0.48 = −0.98

In ONE HUNDRED POINTS, your position went from behaving like
half a short futures lot to behaving like a FULL short futures
lot. You did not change anything. The market changed your
position for you.

Index moves up another 100 points to 25,200.
  Delta is now pinned near −1.00. You are fully short.
  Each further point costs you 75 rupees, with no decay left
  to offset it.

P&L on that 200-point move:
  Premium went from 42.00 to roughly 205.00
  Loss = (205 − 42) x 75 = −Rs 12,225
  against a maximum possible gain of 42 x 75 = Rs 3,150.

A 0.8% index move on expiry day turned a Rs 3,150 maximum gain
into a Rs 12,225 loss. That is gamma.

THE RULE

  Long gamma = the market's moves help you (you get longer as

```
                it rises, shorter as it falls). You pay theta
                for this privilege.
  Short gamma = the market's moves hurt you, increasingly, in
                both directions. You are PAID theta to accept it.

"Collect the theta, ignore the gamma" is the single most common
way Indian retail option sellers blow up on expiry day.
```

Theta

TEXT

THETA — TIME DECAY

Theta is quoted as the premium lost per CALENDAR DAY, all else equal.
For a long option it is always negative. Always.

WORKED — the decay curve of an ATM option

Index 25,000, strike 25,000 CE, IV 14% held constant.

Days to expiry	Premium (points)	Theta (pts/day)	Premium lost that day
30	282.00	-4.60	-
25	258.50	-5.05	-4.70
20	232.00	-5.70	-5.30
15	202.00	-6.70	-6.00
10	164.00	-8.30	-7.60
7	137.00	-10.00	-9.00
5	115.50	-12.00	-10.75
3	89.00	-16.00	-13.25
2	72.50	-21.00	-16.50
1	51.00	-34.00	-21.50
0	0.00	-	-51.00

NOTICE THE SHAPE. The premium does not decay in a straight line.
It decays in proportion to sqrt(time). Roughly:

> Premium is proportional to sqrt(t)
>
> Half the time remaining → only 29% of the premium is gone
> (sqrt(0.5) = 0.707, so 70.7% of value remains)
>
> Last 7 days of a 30-day option: you lose 137 of 282 points,
> i.e. 49% of the value in the last 23% of the time.
>
> Last 2 days: 72.50 points, 26% of the original value in the
> last 6.7% of the time.

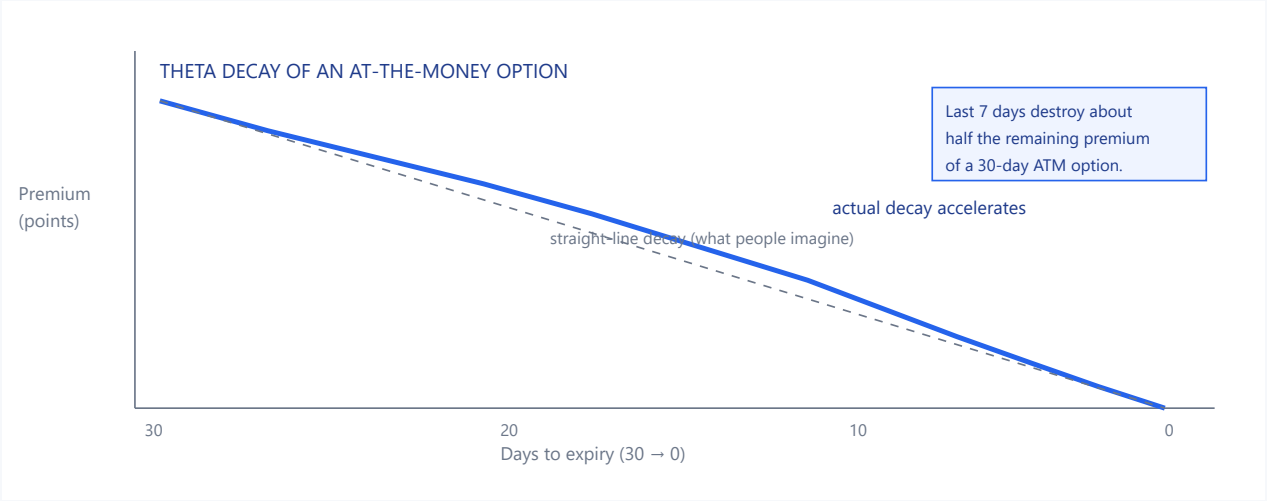
TWO OPPOSITE CONCLUSIONS, BOTH POPULAR, BOTH INCOMPLETE

BUYER'S TRAP: "Buy long-dated, decay is slow." True, but
long-dated options are expensive, have low gamma, and tie up
capital. You need a bigger move to profit.

SELLER'S TRAP: "Sell short-dated, decay is fastest." Also true,
and it is exactly where gamma is highest. The accelerating
theta you want and the accelerating gamma that kills you are
THE SAME PHENOMENON viewed from two angles. You cannot have

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  one without the other. There is no arrangement of strikes and
  expiries that gives you fast decay with low gamma risk.

Figures illustrative. As at late 2026.
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Time value decays roughly with the square root of time remaining, not linearly. The last week is brutal for buyers — and that same week is where gamma risk for sellers is highest.

Vega

TEXT

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VEGA - SENSITIVITY TO IMPLIED VOLATILITY
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Vega = premium change for a 1 PERCENTAGE POINT change in IV.
Long options are always long vega. Short options always short.

WHERE VEGA LIVES
  - Highest: at the money, LONG time to expiry
  - Lowest : deep ITM/OTM, or very close to expiry

Index 25,000, IV 14%. Vega per 1 point of IV:

Strike  Type   45 days   18 days   7 days   1 day
──────  ────  ────────  ────────  ───────  ───────
25,000  CE      22.50     14.20      8.80     3.30
25,300  CE      19.80     10.60      5.10     0.95
25,900  CE      11.20      3.40      0.70     0.03

WORKED - A VEGA DISASTER
────────────────────────
Position: SHORT STRANGLE
  SELL 1 lot 25,600 CE @ 34.50
  SELL 1 lot 24,400 PE @ 46.00
  Total premium received = (34.50 + 46.00) x 75 = Rs 6,037.50
  18 days to expiry. IV = 14%.
  Combined vega (short) = -(3.40 + 4.10) = -7.50 per IV point

OVERNIGHT: a global risk event. The index opens DOWN only 0.6%
to 24,850 - a small move. But IV jumps from 14% to 24%.

Change in premium from VEGA alone:
  IV up 10 points x vega 7.50 = 75.00 points against you
  75.00 x 75 = -Rs 5,625

Change from DELTA (index down 150 points):
  Call delta +0.13 (short, so helps): +0.13 x 150 =  +19.5 pts
  Put  delta -0.14 (short, so hurts): -0.14 x -150 = -21.0 pts
  Net roughly flat.

NET: you have lost almost your entire maximum possible profit
(Rs 6,038) on a 0.6% index move, because VOLATILITY repriced.

Worse: your margin requirement just went UP, because SPAN
margin scales with volatility. You may face a margin call on a
position that is only slightly out of the money.

THE LESSON
  Short-premium positions are short volatility. You can be
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  completely right about direction and lose on vega. For a
  retail account with no vega hedge, this is an unmanaged risk.
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Rho

TEXT

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RHO — SENSITIVITY TO INTEREST RATES
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Rho = premium change for a 1 percentage point change in r.

Index 25,000, 18 days, IV 14%:
  25,000 CE rho = +0.63   (per 1% rate change)
  25,000 PE rho = −0.60

A 25 basis point policy rate change moves the ATM call premium
by about 0.63 x 0.25 = 0.16 points = Rs 12 on a 75 lot.

CONCLUSION: for short-dated Indian equity options, RHO IS NOISE.
It matters for LEAPS-style long-dated options and for interest
rate derivatives. For a weekly or monthly index option it is
below the bid-ask spread. Know what it is; do not think about it.
```

How the Greeks move with moneyness and time

Greek	Deep OTM	At the money	Deep ITM	As expiry approaches
Delta (call)	Near 0	Near 0.5	Near 1	Polarises towards 0 or 1 — the curve becomes a step function
Gamma	Low	**Maximum**	Low	Explodes at the money; collapses everywhere else
Theta	Small (little value left to lose)	**Maximum**	Small	Accelerates sharply in the final week
Vega	Low	**Maximum**	Low	Falls towards zero — an expiring option has nothing left for vol to affect
Rho	Low	Moderate	Highest	Falls towards zero

⚠ **Gamma, theta and vega all peak at the money**

That is not a coincidence — they are three views of the same underlying quantity: how much optionality the contract contains. At-the-money options have the most optionality and therefore the most of all three sensitivities. There is no strike that gives you high theta with low gamma, and no strike that gives you high gamma with low theta cost. Every option strategy is a trade-off within that constraint.

Position Greeks for a multi-leg structure

TEXT

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AGGREGATING GREEKS ACROSS A FOUR-LEG POSITION
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Structure: IRON CONDOR on a hypothetical index
Index spot 25,000, 18 days to expiry, IV 14%, lot size 75

LEG                      Qty  Prem   Delta   Gamma    Theta   Vega
─────────────────────    ───  ─────  ──────  ───────  ──────  ─────
SELL 25,600 CE            -1  34.50  -0.13   -0.00021  +3.10  -3.40
BUY  25,900 CE            +1  10.20  +0.05   +0.00011  -1.30  +1.70
SELL 24,400 PE            -1  46.00  +0.14   -0.00023  +3.60  -4.10
BUY  24,100 PE            +1  16.80  -0.06   +0.00013  -1.60  +2.00
─────────────────────    ───  ─────  ──────  ───────  ──────  ─────
NET POSITION (per unit)         53.50 +0.00  -0.00020  +3.80  -3.80

NET CREDIT RECEIVED
  (34.50 - 10.20 + 46.00 - 16.80) = 53.50 points
  53.50 x 75 = Rs 4,012.50

NOW READ THE NET GREEKS AS A SENTENCE
─────────────────────────────────────

Delta  0.00  → "I have no directional view right now."
Gamma -0.00020 → "If the index MOVES, I get hurt, and the
                  further it moves the faster I get hurt."
Theta +3.80  → "Each day that passes earns me 3.80 x 75
                = Rs 285, if nothing else changes."
Vega  -3.80  → "If IV rises 1 point, I lose 3.80 x 75 = Rs 285."

RUPEE SENSITIVITIES PER UNIT MOVE
  1 day passing              : +Rs    285
  IV up 1 point              : -Rs    285
  IV up 6 points (event)     : -Rs  1,710
  Index moves 300 points     : delta goes from 0.00 to roughly
                               -0.06 (gamma 0.0002 x 300); the
                               position starts acquiring an
                               unwanted directional exposure
  Index moves 600 points     : one short leg is at the money;
                               gamma and vega both spike; the
                               position is now a different trade
                               from the one you put on

BREAK-EVENS AT EXPIRY
  Upper = 25,600 + 53.50 = 25,653.50
  Lower = 24,400 - 53.50 = 24,346.50
  Max profit = Rs 4,012.50 (index between 24,400 and 25,600)
  Max loss   = (300 wing width - 53.50 credit) x 75
             = 246.50 x 75 = Rs 18,487.50

RISK/REWARD: risking Rs 18,487 to make Rs 4,012. Ratio 4.6 : 1
```

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against you. You need to win roughly 82% of the time just to
break even BEFORE transaction costs on eight legs (four in,
four out). Include costs and the required win rate rises further.

Greeks and premiums illustrative. As at late 2026.
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✦ **The one habit that separates serious option traders from gamblers**

Before entering any multi-leg position, write out the net delta, gamma, theta and vega, convert each into rupees, and then write one sentence describing what you are actually betting on. If the sentence is "I am betting the index stays inside a range and volatility does not rise", you now know you have two ways to lose and one way to win. Most retail traders enter iron condors without ever converting vega into rupees.

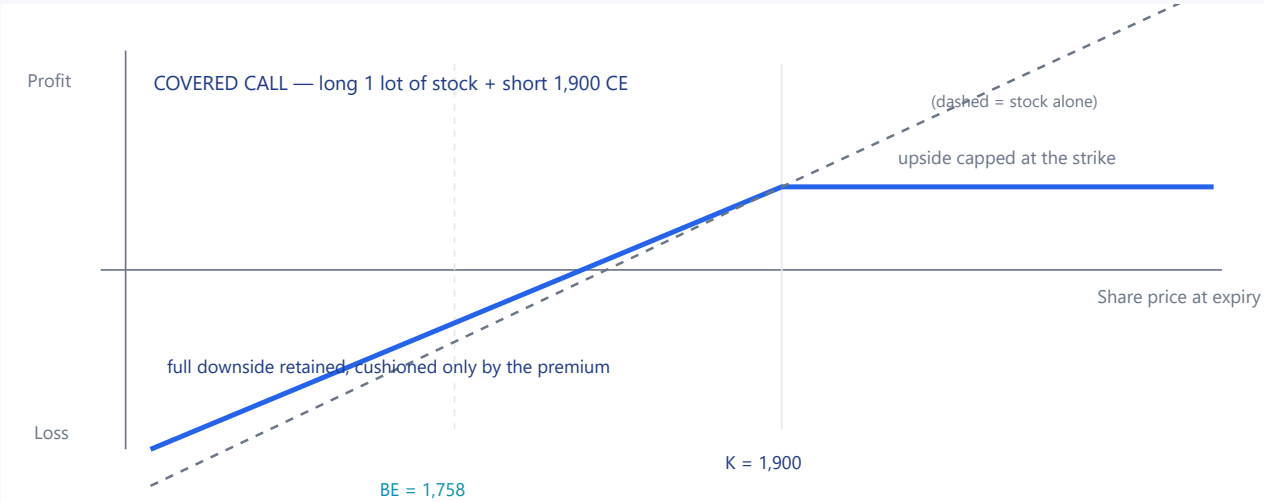
Core option strategies, with the hidden risk stated

Thirteen structures follow. Each gets a construction, worked maximum profit and loss, break-evens, the circumstance in which it is defensible, and — the part usually omitted — the risk that is not visible on the payoff diagram. All examples use a hypothetical index at 25,000 with a lot size of 75 and 18 days to expiry, except where a single stock is required.

⚠ **A strategy is not an edge**

Knowing the construction of an iron condor is like knowing the rules of chess. It tells you nothing about whether you should play, against whom, or at what stake. Every structure below has a negative expected value for a participant with no edge, once costs are included. The structures are tools for *shaping* risk, not for creating returns.

1. Covered call



Covered call. You have sold the right-hand tail of your own stock position in exchange for a fixed cash payment. The left tail is untouched.

TEXT

COVERED CALL - WORKED
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This one needs a single stock, because you must OWN the shares.
Hypothetical: a share trading at Rs 1,800, F&O lot size 500.

POSITION
  You already own 500 shares at an average cost of Rs 1,800
  Holding value = 500 x 1,800 = Rs 9,00,000
  SELL 1 lot of the 1,900 CE @ Rs 42.00
  Premium received = 42 x 500 = Rs 21,000

NUMBERS
  Break-even on the combined position = 1,800 - 42 = Rs 1,758
  Max profit = (1,900 - 1,800 + 42) x 500 = 142 x 500 = Rs 71,000
               (achieved at any price at or above 1,900)
  Max loss   = (1,800 - 42) x 500 = Rs 8,79,000 (share to zero)

PAYOFF TABLE

Share at expiry	Stock P&L (Rs)	Option P&L (Rs)	Total P&L (Rs)	Return on Rs 9,00,000	
1,400	-2,00,000	+21,000	-1,79,000	-19.9%	
1,600	-1,00,000	+21,000	-79,000	-8.8%	
1,758	-21,000	+21,000	0	0.0%	BE
1,800	0	+21,000	+21,000	+2.3%	
1,900	+50,000	+21,000	+71,000	+7.9%	
2,000	+1,00,000	-29,000	+71,000	+7.9%	
2,200	+2,00,000	-1,29,000	+71,000	+7.9%	
2,500	+3,50,000	-2,79,000	+71,000	+7.9%	

WHEN IT IS DEFENSIBLE
  - You own the shares, intend to keep owning them, and would be
    genuinely happy to sell at 1,900.
  - You expect the share to drift sideways or rise modestly.
  - You accept that a 39% rally (to 2,500) earns you the same
    Rs 71,000 as a 5.6% rally.

THE HIDDEN RISKS
  1. OPPORTUNITY LOSS IS REAL LOSS. At 2,500 you forgo
     Rs 2,79,000. "I still made money" is the wrong frame -
     you sold a Rs 2.79 lakh outcome for Rs 21,000.
  2. YOU STILL OWN THE ENTIRE DOWNSIDE. At 1,400 you are down
     Rs 1.79 lakh. The Rs 21,000 premium covered 2.3% of a
     22% fall. A covered call is NOT a hedge.
  3. PHYSICAL SETTLEMENT. Indian single-stock options settle by
     physical delivery. If the share closes above 1,900 on
     expiry, your 500 shares WILL be delivered out. You need
     them in the demat account, unencumbered, and you will face

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     delivery margins in the expiry week (see the margin section).
  4. TAX EVENT. Delivery triggers a capital gains event on the
     shares at a price you did not choose.
  5. YOU CANNOT SELL THE SHARES while short the call without
     converting the position into a naked short call.

Figures illustrative. As at late 2026 - verify lot sizes,
margin and settlement rules at nseindia.com.
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2. Protective put

TEXT

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PROTECTIVE PUT - WORKED
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Long the underlying + long a put. The textbook hedge.

POSITION (index version, for portfolio hedging)
  Portfolio value                 Rs 10,00,000
  Index spot                      25,000
  BUY 1 lot 24,400 PE @ 46.00  (about 2.4% OTM)
  Cost = 46.00 x 75 = Rs 3,450

  Note the notional mismatch: one 24,400 put covers
  24,400 x 75 = Rs 18.3 lakh of notional against a Rs 10 lakh
  portfolio. You are over-hedged by about 83%.
  This is the lot-size granularity problem again.

SCENARIO TABLE (assume portfolio beta 1.0 for clarity)
Index     Portfolio   Put payoff      Put cost    Net
level     P&L         (gross, Rs)     (Rs)        (Rs)
_______  __________  _____________  _________  _________

26,000     +40,000             0      -3,450      +36,550
25,000           0             0      -3,450       -3,450
24,400     -24,000             0      -3,450      -27,450
24,000     -40,000        +30,000     -3,450      -13,450
23,000     -80,000     +1,05,000      -3,450      +21,550
22,000   -1,20,000     +1,80,000      -3,450      +56,550
20,000   -2,00,000     +3,30,000      -3,450    +1,26,550

READ THIS CAREFULLY
  1. Between 25,000 and 24,400 the put does NOTHING. You have a
     2.4% deductible. The worst net outcome is -Rs 27,450, at
     exactly the strike.
  2. Below 24,400 the OVER-HEDGE flips you net positive in a
     crash. That is not a hedge; that is a short position with
     extra steps. A correctly sized hedge would leave you flat.
  3. The Rs 3,450 cost is 0.35% of the portfolio for 18 days.
     Annualised, running this continuously costs roughly 7% a
     year - which will exceed the long-run equity risk premium
     you are trying to protect.

WHEN IT IS DEFENSIBLE
  - A SPECIFIC, TIME-BOXED risk you want to cross: a results
    season, a budget, a known binary event, a period when you
    need the capital intact for a real-world reason.
  - NOT as a permanent overlay. Permanent put buying is a
    reliable way to underperform.

HIDDEN RISKS
  - You pay the volatility skew: OTM puts carry the chain's
```

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    highest IV. You are buying the most expensive insurance
    in the market, by design.
  - Basis and lot granularity mean the hedge is imperfect.
  - If the market drifts sideways you lose the premium with
    nothing to show for it - which is the hedge WORKING.
```

3. Collar

TEXT

COLLAR - WORKED
═══════════════

Long underlying + long OTM put + short OTM call.
You buy the downside protection and PAY for it by selling
the upside. Often structured to be "zero cost".

POSITION (single stock, share at Rs 1,800, lot 500)
  Own 500 shares at Rs 1,800
  BUY 1,700 PE @ Rs 38.00 → cost 38 x 500 = Rs 19,000
  SELL 1,950 CE @ Rs 34.00 → credit 34 x 500 = Rs 17,000
  NET COST = Rs 2,000 (near zero cost)

NUMBERS
  Floor (protected below) = 1,700
  Cap   (capped above)   = 1,950
  Net cost per share   = Rs 4.00
  Max loss = (1,800 - 1,700 + 4) x 500 = 104 x 500 = Rs 52,000
  Max gain = (1,950 - 1,800 - 4) x 500 = 146 x 500 = Rs 73,000

PAYOFF TABLE

Share at expiry	Stock P&L (Rs)	Put P&L (Rs)	Call P&L (Rs)	Total (Rs)
1,200	-3,00,000	+2,31,000	+17,000	-52,000
1,500	-1,50,000	+81,000	+17,000	-52,000
1,700	-50,000	-19,000	+17,000	-52,000
1,800	0	-19,000	+17,000	-2,000
1,900	+50,000	-19,000	+17,000	+48,000
1,950	+75,000	-19,000	+17,000	+73,000
2,400	+3,00,000	-19,000	-2,08,000	+73,000

WHEN IT IS DEFENSIBLE
  - A concentrated holding you cannot or will not sell (ESOPs
    past vesting, a legacy position with a large embedded gain)
    and you want to box the outcome for a defined period.
  - You have a target exit price anyway, and a pain threshold.

HIDDEN RISKS
  - You have sold your upside. In a sustained rally this is
    expensive and psychologically brutal.
  - Both legs are physically settled on Indian single stocks.
    Assignment on the call delivers your shares out; exercise
    of the put delivers them out too. Plan for both.
  - "Zero cost" is marketing. The cost is the forgone upside,
    which in a bull market dwarfs the premium saved.
  - Skew works against you: the put you buy has higher IV than
    the call you sell, so a strictly symmetric collar around
    spot is usually a net DEBIT, not zero.

4. Cash-secured put

TEXT

```
CASH-SECURED PUT - WORKED
=========================
Sell a put AND set aside the full cash to buy the underlying
if assigned. The collateral is what distinguishes this from a
naked short put.

POSITION (single stock at Rs 1,800, lot 500)
  You want to own this share, but at Rs 1,700, not Rs 1,800.
  SELL 1 lot 1,700 PE @ Rs 38.00
  Premium received = Rs 19,000
  CASH SET ASIDE = 1,700 x 500 = Rs 8,50,000

  (If you do not set this aside, you are running a leveraged
   short put, not a cash-secured put. The margin blocked might
   be only Rs 1.4 lakh - that is the point of the distinction.)

OUTCOMES
  Share > 1,700 at expiry : put expires worthless.
    You keep Rs 19,000. Return on Rs 8,50,000 = 2.24% in 18 days.
    Annualised that looks like 45%. IGNORE THAT NUMBER - you
    cannot repeat it reliably and the tail is what matters.

  Share < 1,700 at expiry : you are ASSIGNED.
    You buy 500 shares at Rs 1,700 = Rs 8,50,000 outflow.
    Effective cost = 1,700 - 38 = Rs 1,662 per share.
    Physical delivery - the shares land in your demat account.

SCENARIO TABLE
Share at   Assigned?   Effective       P&L on the
expiry                 position        position (Rs)
_______   _________   _____________   _____________
 2,000     No          cash + 19,000         +19,000
 1,800     No          cash + 19,000         +19,000
 1,700     Borderline  cash + 19,000         +19,000
 1,662     Yes         500 shares @1662              0
 1,500     Yes         500 shares @1662        -81,000
 1,200     Yes         500 shares @1662      -2,31,000
   800     Yes         500 shares @1662      -4,31,000

WHEN IT IS DEFENSIBLE
  - You genuinely want the shares at the strike, you have the
    cash, and you would otherwise have placed a limit buy order
    at that price. The premium is payment for the commitment.

HIDDEN RISKS
  - THE REASON IT FALLS TO 1,500 IS USUALLY A REASON. You are
    systematically buying assets precisely when the news is bad.
  - Your Rs 8.5 lakh is immobilised for 18 days earning the
```

```
  premium only.
- Physical delivery means delivery margins in expiry week and
  an actual stock position you must then manage.
- This strategy has a long left tail and a capped right tail.
  A sequence of 2.24% wins does not compensate for one −27%
  assignment on a stock that keeps falling.
```

5. Bull call spread (debit vertical)



Bull call spread. Both the loss and the gain are boxed. You have sold away the far upside to reduce the cost of entry.

TEXT

```
BULL CALL SPREAD - WORKED
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BUY  1 lot 25,000 CE @ 218.50   → debit  Rs 16,387.50
SELL 1 lot 25,300 CE @  96.25   → credit Rs  7,218.75
NET DEBIT = 122.25 points      = Rs  9,168.75

NUMBERS
  Spread width   = 25,300 - 25,000 = 300 points
  Max profit     = (300 - 122.25) x 75 = 177.75 x 75 = Rs 13,331.25
  Max loss       = net debit = Rs 9,168.75
  Break-even     = 25,000 + 122.25 = 25,122.25
  Risk : reward  = 9,168.75 : 13,331.25 = 1 : 1.45

PAYOFF TABLE
Index at   Long 25,000  Short 25,300   Net per    Total
expiry     CE value     CE value       unit       P&L (Rs)
_______    __________   ____________   ________   __________
24,500          0.00         0.00      -122.25    -9,168.75
25,000          0.00         0.00      -122.25    -9,168.75
25,100        100.00         0.00       -22.25    -1,668.75
25,122.25     122.25         0.00         0.00         0.00
25,200        200.00         0.00       +77.75    +5,831.25
25,300        300.00         0.00      +177.75   +13,331.25
25,600        600.00      -300.00      +177.75   +13,331.25
26,500      1,500.00    -1,200.00      +177.75   +13,331.25

COMPARE WITH JUST BUYING THE 25,000 CE OUTRIGHT
Index at    Spread P&L    Naked long call P&L
expiry      (Rs)          (Rs)
_______     __________    ___________________
24,500       -9,168.75          -16,387.50
25,000       -9,168.75          -16,387.50
25,300      +13,331.25           +6,112.50
25,600      +13,331.25          +28,612.50
26,500      +13,331.25          +96,112.50

THE TRADE-OFF, PLAINLY
  The spread loses LESS when wrong (Rs 9,169 vs Rs 16,388)
  and earns MORE for a modest move (to 25,300).
  It earns far less for a large move (Rs 13,331 vs Rs 96,113
  at 26,500).

You have sold the tail. If your view is "up a bit", the spread
is the better expression. If your view is "up enormously", the
spread caps exactly the outcome you were trying to capture.

WHEN IT IS DEFENSIBLE
  - You have a bounded directional view with a target price.
```

```
  - You want defined risk and a lower break-even than a naked
    long call.
  - IV is elevated: the short leg recovers some of the inflated
    premium you are paying on the long leg.

HIDDEN RISKS
  - TWO legs in, TWO legs out = four sets of charges. On a
    Rs 9,169 position that cost is a meaningful percentage.
  - Liquidity: the far strike may have a wider spread; you may
    not get the mid price on both legs.
  - If the index closes between 25,000 and 25,300 at expiry,
    the long leg is ITM and the short is not. For INDEX options
    this is cash-settled and harmless. For STOCK options it
    creates a physical delivery obligation on one leg only -
    a very expensive surprise. Never hold a stock vertical
    through expiry without understanding this.
```

6. Bear put spread (debit vertical)

TEXT

```
BEAR PUT SPREAD - WORKED
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The mirror of the bull call spread, for a bounded bearish view.

BUY  1 lot 25,000 PE @ 196.00   → debit  Rs 14,700.00
SELL 1 lot 24,700 PE @  82.75   → credit Rs  6,206.25
NET DEBIT = 113.25 points      = Rs  8,493.75

NUMBERS
  Spread width  = 300 points
  Max profit    = (300 - 113.25) x 75 = 186.75 x 75 = Rs 14,006.25
  Max loss      = Rs 8,493.75
  Break-even    = 25,000 - 113.25 = 24,886.75
  Risk : reward = 1 : 1.65

PAYOFF TABLE
Index at   Net per unit   Total P&L (Rs)
────────   ────────────   ──────────────
25,500        -113.25         -8,493.75
25,000        -113.25         -8,493.75
24,900          -13.25         -  993.75
24,886.75         0.00              0.00
24,800         +86.75         +6,506.25
24,700        +186.75        +14,006.25
24,000        +186.75        +14,006.25
23,000        +186.75        +14,006.25

WHEN IT IS DEFENSIBLE
  - A defined bearish view over a defined window, with capped
    risk. This is the responsible alternative to shorting
    futures, which has unbounded loss.
  - As a replacement for a stop-loss on a long position: see
    the legitimate-use-cases section.

HIDDEN RISKS
  - You are BUYING the high-IV side of the skew on the long leg.
    Bear put spreads are structurally expensive in equity indices.
  - Same four-leg cost and stock-delivery considerations as the
    bull call spread.
  - Markets drift up over long horizons. Persistent bearish
    spread buying has a structural headwind beyond the premium.
```

7. Long straddle

TEXT

LONG STRADDLE - WORKED
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BUY the ATM call AND the ATM put. A pure bet on MOVEMENT,
in either direction, with no directional view.

BUY 1 lot 25,000 CE @ 218.50 → Rs 16,387.50
BUY 1 lot 25,000 PE @ 196.00 → Rs 14,700.00
TOTAL DEBIT = 414.50 points   = Rs 31,087.50

NUMBERS
  Max loss   = Rs 31,087.50 (index closes exactly at 25,000)
  Upper BE   = 25,000 + 414.50 = 25,414.50   (+1.66%)
  Lower BE   = 25,000 - 414.50 = 24,585.50   (-1.66%)
  Max profit   = unlimited upside; very large downside

PAYOFF TABLE

Index at	Call value	Put value	Net/unit	Total P&L (Rs)
23,500	0.00	1,500.00	+1,085.50	+81,412.50
24,000	0.00	1,000.00	+585.50	+43,912.50
24,585.50	0.00	414.50	0.00	0.00
24,800	0.00	200.00	-214.50	-16,087.50
25,000	0.00	0.00	-414.50	-31,087.50
25,200	200.00	0.00	-214.50	-16,087.50
25,414.50	414.50	0.00	0.00	0.00
26,000	1,000.00	0.00	+585.50	+43,912.50
26,500	1,500.00	0.00	+1,085.50	+81,412.50

THE PROBABILITY CHECK - DO THIS EVERY TIME
──

Expected 1-SD move at IV 14%, 18 days = 777 points (computed
earlier). Your break-evens are at +/- 414.50 points, which is
414.50 / 777 = 0.53 standard deviations.

Probability of finishing OUTSIDE +/- 0.53 SD is roughly 60%.
So this looks favourable? NO - because that 60% includes closes
only just beyond the break-even, where you make almost nothing.

The honest framing: the straddle is priced so that the EXPECTED
profit is approximately zero before costs and NEGATIVE after.
You are not getting a 60% edge. You are paying for a move the
market has already priced.

WHEN IT IS DEFENSIBLE
  - You believe realised volatility will EXCEED implied
    volatility - i.e. you think 777 points is too low an
    estimate of the coming move. That is a volatility view,
    not a direction view, and it is a hard view to hold well.

```
  - BEFORE an event where IV has NOT yet risen. Buying a
    straddle after IV has already inflated is the IV-crush trap.

HIDDEN RISKS
  - DOUBLE THETA. You are paying decay on two options. At 18
    days that is roughly (5.90 + 5.40) x 75 = Rs 848 per day.
    Fourteen flat days cost you Rs 11,872 - 38% of the premium
    - with the index unchanged.
  - DOUBLE VEGA, which cuts both ways. If IV falls 3 points you
    lose roughly (14.20 + 13.60) x 3 x 75 = Rs 6,255.
  - You need the move to be BIG and SOON. Big but slow is a loss.
```

8. Long strangle

TEXT

```
LONG STRANGLE - WORKED
======================
Same idea as a straddle, but using OTM strikes. Cheaper to
enter, further to travel.

BUY 1 lot 25,300 CE @ 96.25  --> Rs 7,218.75
BUY 1 lot 24,700 PE @ 82.75  --> Rs 6,206.25
TOTAL DEBIT = 179.00 points  = Rs 13,425.00

NUMBERS
  Max loss  = Rs 13,425 (index anywhere between 24,700-25,300)
  Upper BE  = 25,300 + 179.00 = 25,479.00  (+1.92%)
  Lower BE  = 24,700 - 179.00 = 24,521.00  (-1.92%)

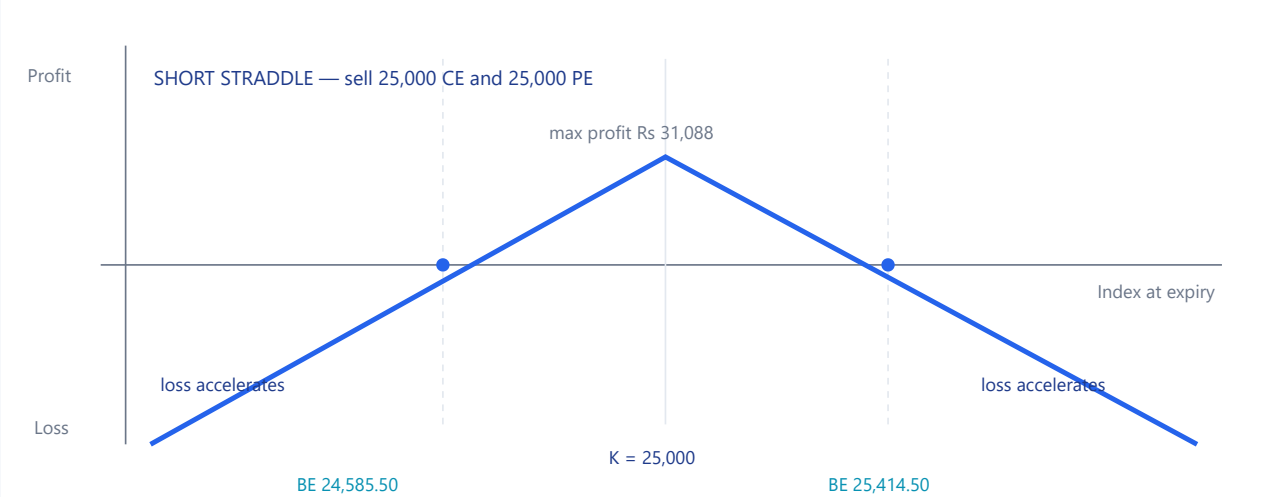
STRADDLE vs STRANGLE, SIDE BY SIDE
                         Straddle      Strangle
  Cost                   Rs 31,088     Rs 13,425
  Upper break-even       25,414.50     25,479.00
  Lower break-even       24,585.50     24,521.00
  Width of the loss zone  single pt    600 points
  Max-loss zone          exactly       the ENTIRE range
                         25,000        24,700-25,300
  P&L at 26,000          +43,913       +56,575
  P&L at 25,000          -31,088       -13,425
  P&L at 25,200          -16,088       -13,425

READ THAT TABLE
  The strangle costs 57% less and pays MORE on a big move.
  It also loses its FULL premium across a 600-point band rather
  than at one exact point.
  In practice, for a given rupee budget you can buy about 2.3x
  as many strangles as straddles - which is how a position that
  "costs less" ends up being a LARGER risk, because people
  size by premium rather than by exposure.

WHEN IT IS DEFENSIBLE
  Same conditions as the straddle, when you want a cheaper
  expression and expect a very large move.

HIDDEN RISKS
  - Probability of finishing outside +/- 1.92% (0.74 SD) in 18
    days is roughly 46%. Probability of finishing far enough
    outside to make a worthwhile return is considerably lower.
  - Pinning: markets often close near large open-interest
    strikes on expiry. Your 600-point dead zone is where the
    index spends most of its time.
```

9. Short straddle



Short straddle. A narrow tent of profit flanked by two lines that keep descending off the page. The chart edge is not a floor.

TEXT

```
SHORT STRADDLE - WORKED
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SELL 1 lot 25,000 CE @ 218.50  → Rs 16,387.50 received
SELL 1 lot 25,000 PE @ 196.00  → Rs 14,700.00 received
TOTAL CREDIT = 414.50 points   = Rs 31,087.50
Margin blocked (illustrative): Rs 2,20,000 - 3,00,000

NUMBERS
  Max profit  = Rs 31,087.50 (index closes EXACTLY at 25,000)
  Upper BE    = 25,414.50
  Lower BE    = 24,585.50
  Max loss    = UNLIMITED on the upside, enormous on the downside

PAYOFF TABLE
Index at    Net/unit     Total P&L (Rs)   Return on
expiry                                     Rs 2,60,000 margin
_______    _________    _____________   __________________
25,000       +414.50        +31,087.50            +11.96%
25,200       +214.50        +16,087.50             +6.19%
25,414.50       0.00              0.00              0.00%
25,600       -185.50        -13,912.50             -5.35%
26,000       -585.50        -43,912.50            -16.89%
26,500     -1,085.50        -81,412.50            -31.31%
27,000     -1,585.50      -1,18,912.50            -45.74%
28,000     -2,585.50      -1,93,912.50            -74.58%
23,000     -1,585.50      -1,18,912.50            -45.74%
22,000     -2,585.50      -1,93,912.50            -74.58%
20,000     -4,585.50      -3,43,912.50           -132.27%

THE PITCH AND THE ARITHMETIC
____________________________
THE PITCH: "The index stays within 1.66% about 70% of the time.
Sell the straddle every cycle and collect Rs 31,088. Twelve
cycles a year = Rs 3.7 lakh on Rs 2.6 lakh of margin."

THE ARITHMETIC: run it as a 12-cycle sequence where 10 cycles
land in the profit zone and 2 do not.

  10 winning cycles, average profit 60% of max:
      10 x 18,653 =  +Rs 1,86,530
   1 moderate loss (index moves 3%):
       1 x -43,913 =   -Rs 43,913
   1 bad cycle (index moves 7% - happens most years):
       1 x -1,93,913 = -Rs 1,93,913
  ____________________________________________________
  TWELVE-MONTH TOTAL                 = -Rs 51,296

  Plus 48 legs of transaction costs (12 cycles x 4 legs),
```

```
  roughly Rs 350 per leg all-in on this size = -Rs 16,800
  ──────────────────────────────────────────────────────

  NET                                  = -Rs 68,096

  On Rs 2.6 lakh of margin that is -26% for a year of work
  in which you were RIGHT 83% OF THE TIME.

THIS IS THE CORE OF WHY ~90% LOSE.
A high win rate with an unbounded loss is not an edge.
It is a loan from the market that gets called in.

HIDDEN RISKS BEYOND THE PAYOFF TABLE
  1. VEGA. A volatility spike loses you money with no index
     move at all. See the vega worked example earlier.
  2. MARGIN EXPANSION. When volatility rises, SPAN margin rises.
     Your position moves against you AND your margin requirement
     increases simultaneously. You can be forced to close at the
     worst price because you cannot fund the margin.
  3. GAP RISK. Overnight gaps bypass every stop loss you set.
     There is no stop that protects a short straddle from a
     7% gap open.
  4. GAMMA on expiry day. See the gamma worked example.
  5. SEBI's additional expiry-day tail-risk margin on short
     options exists precisely because of this risk profile.
     Verify current requirements at nseindia.com.
```

Short straddles and strangles are the single most dangerous retail structure

They are popular because they win frequently and the losses arrive rarely. That combination is maximally seductive and maximally destructive. A retail account with no vega hedge, no intraday risk system, no ability to trade through a gap, and a margin requirement that expands exactly when the position is losing, has no business holding an uncovered short straddle. If you take one instruction from this guide, take this one.

10. Short strangle

TEXT

```
SHORT STRANGLE - WORKED
═══════════════════════
SELL 1 lot 25,600 CE @ 34.50 → Rs 2,587.50
SELL 1 lot 24,400 PE @ 46.00 → Rs 3,450.00
TOTAL CREDIT = 80.50 points  = Rs 6,037.50
Margin blocked (illustrative): Rs 2,00,000 - 2,70,000

NUMBERS
  Max profit = Rs 6,037.50
  Upper BE   = 25,600 + 80.50 = 25,680.50   (+2.72%)
  Lower BE   = 24,400 - 80.50 = 24,319.50   (-2.72%)
  Max loss   = unlimited / enormous

THE SEDUCTION, QUANTIFIED
  Probability of the index finishing between 24,319 and 25,680
  (roughly +/- 0.875 SD at IV 14%, 18 days) is about 62%...
  but the profit zone where you keep the FULL premium
  (24,400 to 25,600) has a probability of about 86%.

  "An 86% win rate!" - and that is how it is sold.

NOW THE EXPECTED VALUE, PROPERLY
  Outcome                          Prob    P&L (Rs)    Contribution
  ──────────────────────────────   ─────   ──────────  ────────────
  Finishes inside 24,400-25,600    0.86     +6,038       +5,193
  Modest breach (to 1.5 SD)        0.10    -35,000       -3,500
  Large breach (2-3 SD)            0.03   -1,20,000      -3,600
  Tail event (>3 SD)               0.01   -3,00,000      -3,000
  ──────────────────────────────   ─────   ──────────  ────────────
  EXPECTED VALUE PER CYCLE                                -Rs 4,907

  Then subtract four legs of transaction costs: about -Rs 1,000.
  EXPECTED VALUE = roughly -Rs 5,900 PER CYCLE.

  The probabilities above are illustrative, but the SHAPE is
  robust: a structure with an 86% win rate and a 50:1
  loss-to-gain ratio in the tail has a negative expected value
  unless the tail probability is materially lower than the
  option market's own pricing implies. And the option market
  prices the tail for a living.

IF THE MARKET'S PROBABILITY ESTIMATE IS CORRECT, YOUR EV IS
APPROXIMATELY ZERO BEFORE COSTS AND NEGATIVE AFTER.
To profit you must believe the market systematically overprices
the tail. Sometimes it does. You will not know when.
```

11. Iron condor



Iron condor. Defined risk on both wings — but the loss rectangle is several times taller than the profit plateau, which is why the required win rate is so high.

TEXT

```
IRON CONDOR - WORKED
====================
Short strangle + protective wings. The Greeks table for this
exact structure appears in the Greeks section above.

SELL 1 lot 25,600 CE @ 34.50   +Rs 2,587.50
BUY  1 lot 25,900 CE @ 10.20   -Rs   765.00
SELL 1 lot 24,400 PE @ 46.00   +Rs 3,450.00
BUY  1 lot 24,100 PE @ 16.80   -Rs 1,260.00
NET CREDIT = 53.50 points      = Rs 4,012.50

NUMBERS
  Wing width     = 300 points each side
  Max profit     = Rs 4,012.50  (index 24,400 to 25,600)
  Max loss       = (300 - 53.50) x 75 = Rs 18,487.50
  Upper BE       = 25,600 + 53.50 = 25,653.50
  Lower BE       = 24,400 - 53.50 = 24,346.50
  Margin: far lower than a naked strangle because the wings
          cap the risk - typically Rs 20,000-35,000. Verify.

PAYOFF TABLE
Index at    Net/unit   Total P&L (Rs)
________    ________   ______________
23,500       -246.50      -18,487.50
24,100       -246.50      -18,487.50
24,346.50       0.00            0.00
24,400       +53.50       +4,012.50
25,000       +53.50       +4,012.50
25,600       +53.50       +4,012.50
25,653.50       0.00            0.00
25,900       -246.50      -18,487.50
27,000       -246.50      -18,487.50

THE BREAK-EVEN WIN RATE
_______________________
Let p = probability of a full-profit outcome.
Ignore partial outcomes for simplicity.

  EV = p x 4,012.50 - (1-p) x 18,487.50 = 0
  p x 4,012.50 = 18,487.50 - p x 18,487.50
  p x 22,500 = 18,487.50
  p = 0.8217

You must win 82.2% of the time merely to break even,
BEFORE transaction costs on eight legs.

With costs of roughly Rs 1,800 per complete round trip
(8 legs on this size):
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  EV = p x 4,012.50 - (1-p) x 18,487.50 - 1,800 = 0
  p x 22,500 = 20,287.50
  p = 0.9017

YOU MUST WIN MORE THAN 90% OF THE TIME.

Is a 90% win rate achievable? The structure's own probability
of full profit is roughly 86%. The market has priced the credit
so that it is NOT comfortably achievable. This is not a flaw in
the iron condor; it is the market functioning.

WHEN IT IS DEFENSIBLE
  - You have a genuine, evidenced view that implied volatility
    is too high relative to what will be realised.
  - You can manage the position actively and will close early
    rather than holding into gamma risk.
  - You size it so that several consecutive max losses do not
    matter to you.

HIDDEN RISKS
  - EIGHT legs of transaction cost. This is the dominant
    practical problem.
  - The wings are OTM and illiquid; you may not get good fills,
    especially when you need to exit urgently.
  - Managing the position (rolling a tested side) converts the
    defined risk into something else and adds more cost.
  - The temptation to "widen the wings for more credit" is the
    temptation to convert it back into a short strangle.
```

12. Butterfly

TEXT

```
LONG CALL BUTTERFLY - WORKED
============================
BUY  1 lot 24,700 CE @ 382.00   -Rs 28,650.00
SELL 2 lots 25,000 CE @ 218.50  +Rs 32,775.00
BUY  1 lot 25,300 CE @  96.25   -Rs  7,218.75
NET DEBIT = 41.25 points        = Rs  3,093.75

NUMBERS
  Max profit = (300 - 41.25) x 75 = 258.75 x 75 = Rs 19,406.25
               achieved ONLY if the index closes EXACTLY
               at 25,000
  Max loss   = net debit = Rs 3,093.75
  Lower BE   = 24,700 + 41.25 = 24,741.25
  Upper BE   = 25,300 - 41.25 = 25,258.75
  Risk:reward = 1 : 6.27  (the best ratio in this whole section)

PAYOFF TABLE
Index at    Net/unit    Total P&L (Rs)
________    ________    ______________
24,000        -41.25       -3,093.75
24,700        -41.25       -3,093.75
24,741.25       0.00             0.00
24,850      +108.75        +8,156.25
25,000      +258.75       +19,406.25   <- the single peak
25,150      +108.75        +8,156.25
25,258.75       0.00             0.00
25,300        -41.25       -3,093.75
26,000        -41.25       -3,093.75

WHY THE RATIO IS SO GOOD AND WHY IT DOES NOT HELP
  The 6.27:1 payoff exists because the probability of landing
  within a 300-point window AND near its centre is low. The
  market is not giving you anything. You get a great ratio
  precisely because you are unlikely to collect it.

  Probability of finishing within +/- 258.75 points (0.33 SD)
  is roughly 26%. Probability of finishing near the peak is
  far lower still.

  EV check (crude): 0.26 x (an average partial profit of, say,
  Rs 7,000) - 0.74 x 3,093.75 = 1,820 - 2,289 = -Rs 469,
  before four legs of cost.

WHEN IT IS DEFENSIBLE
  - You have a specific pinning view: you believe the index
    will be near a particular level at a particular expiry.
  - You want a very cheap, very defined expression of that.
  - As a low-cost way to be long gamma around a level.
```

```
HIDDEN RISKS
  - FOUR legs (the middle is two contracts) = eight fills
    round trip. Cost as a percentage of a Rs 3,094 position
    is severe.
  - The profit is extremely path-dependent and only crystallises
    at expiry. Mid-cycle the position barely moves.
  - On single stocks, the physical-settlement complexity across
    three strikes is a genuine operational hazard.
```

13. Calendar spread (horizontal)

TEXT

```
CALENDAR SPREAD - WORKED
═══════════════════════

SELL 1 lot NEAR-expiry 25,000 CE  @ 218.50 (18 days) +Rs 16,387.50
BUY  1 lot FAR-expiry  25,000 CE  @ 352.00 (46 days) -Rs 26,400.00
NET DEBIT = 133.50 points                           = Rs 10,012.50

WHAT YOU ARE ACTUALLY TRADING
  Not direction. You are long the DIFFERENCE between the far
  option's slower decay and the near option's faster decay,
  and you are LONG VEGA on the back month.

  Net theta  : POSITIVE (near leg decays faster than far leg)
  Net vega   : POSITIVE (far leg has more vega than near leg)
  Net delta  : near zero at entry
  Net gamma  : NEGATIVE (short the high-gamma near option)

THE UNUSUAL PROPERTY
  This is the only common structure that is simultaneously
  POSITIVE THETA and POSITIVE VEGA. Most structures force you
  to choose. That is why it is used by desks ahead of an
  expected volatility expansion.

OUTCOME AT NEAR-EXPIRY (18 days from now)
Index at    Near CE      Far CE (est,   Net value   P&L
near exp    expires at   28 days left)  per unit    (Rs)
________    __________   _____________  _________   _________
24,000          0.00          22.00        22.00    -8,362.50
24,500          0.00          74.00        74.00    -4,462.50
24,800          0.00         158.00       158.00    +1,837.50
25,000          0.00         212.00       212.00    +5,887.50
25,200        200.00         290.00        90.00    -3,262.50
25,500        500.00         540.00        40.00    -7,012.50
26,000      1,000.00       1,020.00        20.00    -8,512.50

  Max profit occurs with the index AT the strike at near expiry.
  Loss widens in both directions - the opposite shape to the
  straddle and the same shape as the butterfly.

IF IV RISES 4 POINTS WHILE THE INDEX STAYS AT 25,000:
  Far leg vega roughly 20.00, near leg roughly 14.20
  Net vega = +5.80 per IV point
  Gain = 5.80 x 4 x 75 = +Rs 1,740 on top of the theta gain.

WHEN IT IS DEFENSIBLE
  - You expect the underlying to stay near a level in the short
    term AND volatility to rise over the medium term.
  - Post-event, when near-dated IV has crushed but longer-dated
    IV has not.
```

```
HIDDEN RISKS
  - SEBI removed the calendar-spread MARGIN BENEFIT on expiry
    day. Holding this into the near leg's expiry now attracts
    full margin on both legs. Verify current treatment.
  - You are short the near option: if the index moves sharply
    toward expiry, near-leg gamma dominates and the far leg
    does not keep pace.
  - On single stocks, the near leg physically settles while the
    far leg does not - you can end up with an unwanted share
    position and a naked long option.
  - Liquidity in far-month options is much thinner. Your exit
    price may be materially worse than the screen suggests.

All premiums illustrative. As at late 2026 - verify contract
specifications and margin treatment at nseindia.com.
```

Strategy comparison table

Strategy	View	Net debit/credit	Max profit	Max loss	Th
Covered call	Neutral to mildly up	Credit	Capped	Very large (stock to zero)	+
Protective put	Long, want a floor	Debit	Uncapped	Capped	–
Collar	Boxed outcome	Near zero	Capped	Capped	~(
Cash-secured put	Willing buyer lower	Credit	Capped	Very large	+
Bull call spread	Moderately up	Debit	Capped	Capped	–
Bear put spread	Moderately down	Debit	Capped	Capped	–
Long straddle	Big move, no direction	Debit	Uncapped	Capped	–
Long strangle	Very big move	Debit	Uncapped	Capped	–
Short straddle	No move	Credit	Capped	**Unlimited**	+
Short strangle	No move	Credit	Capped	**Unlimited**	+
Iron condor	Range-bound	Credit	Capped	Capped	+

Butterfly	Pin at a level	Debit	Capped	Capped	+
Calendar	Flat now, vol later	Debit	Capped	Capped (roughly)	+

Margins in India: SPAN, exposure and the penalty regime

Margin is not a deposit and it is not a cost. It is collateral the clearing corporation holds against the possibility that you will be unable to meet your obligations. Understanding exactly how it is computed, when it changes, and what happens when you fall short is more important to your survival than any strategy in the previous section.

⚠ **Margin rules in India changed repeatedly between 2020 and 2026**
Peak margin reporting, upfront margin collection, the pledge/re-pledge mechanism for collateral, upfront premium collection from option buyers, and expiry-day tail-risk margin have all been introduced or revised. Every number in this section is illustrative **as at late 2026**. Compute live requirements on the exchange margin calculator at **nseindia.com** and confirm with your broker, who may impose a higher requirement than the exchange minimum.

The components of initial margin

Component	What it covers	Typical magnitude (illustrative)
SPAN margin	The worst-case single-day loss of the whole portfolio, computed by simulating a grid of price and volatility scenarios. Portfolio-based: offsetting legs reduce it.	The bulk of the requirement — often 8–12% of notional for an index future
Exposure margin	An additional buffer over and above SPAN, set by the exchange to cover risks SPAN's scenario grid may miss.	Roughly 2–3% of notional for index positions; higher for single stocks
Extreme loss margin (ELM)	Similar role; terminology varies by segment and exchange.	Included in the above
Additional / tail-risk margin	Imposed in specific circumstances — notably an additional margin on short options on expiry day, introduced in the 2024–25 framework.	An extra slice on short option positions on expiry day; verify current level
Mark-to-market (MTM) margin	Not part of initial margin — this is the actual daily cash settlement of gains and losses on futures.	Whatever the day's move produces
Delivery margin	Levied in the expiry week on in-the-money single-stock F&O positions heading for physical settlement. Ramps up over the final days.	Escalates toward the full value of the delivery obligation

How SPAN actually works

TEXT

```
SPAN - STANDARD PORTFOLIO ANALYSIS OF RISK
==========================================

SPAN does NOT apply a fixed percentage. It simulates your ENTIRE
portfolio across a grid of scenarios and charges you the worst one.

THE SCENARIO GRID (conceptually, 16 scenarios)
  For each combination of:
     price move: 0, +1/3, -1/3, +2/3, -2/3, +3/3, -3/3 of the
                 "price scan range"
     volatility move: up or down by the "volatility scan range"
   ... plus two extreme-move scenarios at a fractional weight.

  Compute the portfolio's P&L in each scenario.
  SPAN margin = the LARGEST LOSS across all scenarios.

WHY THIS MATTERS TO YOU
-----------------------------------------------------------------

1. HEDGED POSITIONS GET MARGIN RELIEF AUTOMATICALLY.
   An iron condor's wings cap the loss in the extreme scenarios,
   so SPAN charges far less than for a naked strangle. You do
   not have to ask for this; SPAN sees the whole portfolio.

2. MARGIN RISES WITH VOLATILITY.
   The price scan range and volatility scan range are derived
   from recent volatility. When the market gets violent, the
   scan ranges widen, and EVERY position's margin goes up -
   including positions you opened weeks ago.

3. THE SEQUENCE THAT KILLS ACCOUNTS
   Market gaps --> your short option loses money (MTM debit)
               --> volatility spikes
               --> SPAN scan range widens
               --> margin requirement INCREASES
               --> you now have less cash AND need more margin
               --> margin shortfall --> penalty --> forced square-off
               --> you realise the loss at the worst price

   Three independent things hit you simultaneously and they are
   CORRELATED BY CONSTRUCTION. This is not bad luck.
```

Margin on buying versus writing options

	Buying an option	Writing (selling) an option
What you pay upfront	The full premium, collected upfront (SEBI mandated upfront collection of option premium from buyers in the 2024–25 framework)	SPAN + exposure margin, which is typically many times the premium received
Ongoing margin	None — your loss is already capped at the premium paid	Yes — recomputed daily and intraday; can increase
MTM calls	No cash calls	Yes, effectively, via margin top-ups
Can you be forced out?	No	Yes, on margin shortfall
Illustrative numbers (25,000 CE)	Pay ₹16,388 and that is the end of it	Receive ₹16,388 and block ₹2,20,000; expect it to move

Why upfront premium collection from buyers was introduced

Previously, brokers could allow intraday option buying against notional or unsettled funds. SEBI's framework requires the premium to be collected upfront from the buyer. The practical effect for retail is that you cannot buy options on credit within the day, which removes one source of uncontrolled intraday leverage. Verify the current requirement at sebi.gov.in.

Fully worked margin requirement for a two-leg position

TEXT

```
WORKED MARGIN - BULL CALL SPREAD vs NAKED SHORT CALL
═══════════════════════════════════════════════════

Index 25,000, lot size 75, notional per lot = Rs 18,75,000

POSITION A: NAKED SHORT 25,000 CE
─────────────────────────────────

  Premium received                          +Rs  16,387.50

  SPAN margin (illustrative, 9.2% of notional)
    18,75,000 x 0.092                     =  Rs 1,72,500.00
  Exposure margin (illustrative, 2.5%)
    18,75,000 x 0.025                     =  Rs   46,875.00
                                              ──────────────
  TOTAL INITIAL MARGIN                     =  Rs 2,19,375.00
  Less premium received (credited)         -  Rs   16,387.50
                                              ──────────────
  NET CASH BLOCKED                         =  Rs 2,02,987.50

  Maximum gain          Rs 16,387.50
  Return on capital if it works     +8.07%
  Maximum loss          UNLIMITED

POSITION B: BULL CALL SPREAD (buy 25,000 CE, sell 25,300 CE)
────────────────────────────────────────────────────────────

  Wait - that is a DEBIT spread; there is no short-option
  margin at all. You pay the net debit and that is the
  entire capital requirement:

  BUY  25,000 CE @ 218.50   -Rs 16,387.50
  SELL 25,300 CE @  96.25   + Rs 7,218.75
  NET DEBIT PAID                           =  Rs  9,168.75
  ADDITIONAL MARGIN                        =  Rs      0.00

  Because the long leg is lower-strike than the short leg,
  the position can never lose more than the debit, and SPAN
  recognises this. Your entire capital at risk is Rs 9,169.

POSITION C: CREDIT SPREAD (sell 25,000 CE, buy 25,300 CE)
─────────────────────────────────────────────────────────

  SELL 25,000 CE @ 218.50  +Rs 16,387.50
  BUY  25,300 CE @  96.25  - Rs  7,218.75
  NET CREDIT                               =  Rs  9,168.75

  SPAN now sees a capped maximum loss of
    (300 - 122.25) x 75 = Rs 13,331.25
  Margin charged (illustrative)            =  Rs 14,500.00
  Less net credit received                 -  Rs  9,168.75
  NET CASH BLOCKED                         =  Rs  5,331.25
```

```
COMPARE THE CAPITAL EFFICIENCY
  Position          Capital    Max gain    Max loss
  ________________  _________  __________  ___________

  A naked short     2,02,988      16,388   UNLIMITED
  B debit spread       9,169      13,331       9,169
  C credit spread      5,331       9,169      13,331

  The naked short requires 22x the capital of the debit
  spread to earn roughly the same maximum gain, AND has
  unlimited downside. SPAN is telling you something.

All margin percentages illustrative. As at late 2026 - compute
live on the exchange margin calculator at nseindia.com.
```

The margin calculator is a risk tool, not an admin tool

Before any trade, enter the proposed position in the exchange or broker margin calculator and look at the number. Then ask: "if the exchange thinks this position could lose ₹2.2 lakh in a single day's worst case, am I comfortable with that?" The margin requirement is a professional risk system's opinion of your trade. Most retail traders treat it as a hurdle to be minimised rather than information to be read.

Peak margin and intraday monitoring

TEXT

```
PEAK MARGIN – WHAT CHANGED AND WHY IT MATTERS
═══════════════════════════════════════════════

HISTORICALLY: margin was checked at END OF DAY. A trader could
run a huge intraday position on minimal margin, square off
before close, and never be measured.

NOW: the clearing corporation takes random SNAPSHOTS during the
session (multiple per day) and the HIGHEST margin requirement
observed across those snapshots is the "peak margin". You must
have had that margin available at that moment.

CONSEQUENCE FOR INTRADAY TRADERS
  The leverage available intraday is now broadly similar to the
  leverage available overnight. The old "20x intraday margin"
  offers largely disappeared.

SEBI'S 2024–25 ADDITION: INTRADAY POSITION LIMIT MONITORING
  Exchanges now monitor derivative position limits intraday
  rather than only at end of day, with a specified number of
  random snapshots. The intent is to prevent participants from
  exceeding limits intraday and unwinding before the end-of-day
  measurement.

Verify current peak-margin methodology, snapshot frequency and
position-limit monitoring rules at sebi.gov.in and nseindia.com.
As at late 2026.
```

Margin shortfall penalties

TEXT

MARGIN SHORTFALL PENALTY - ILLUSTRATIVE STRUCTURE
═══

If the margin available falls below the requirement, the clearing corporation levies a penalty on the SHORTFALL amount, passed through to you by your broker.

Illustrative slab structure (VERIFY CURRENT RATES):

Shortfall size	Penalty rate (per day)
< Rs 1 lakh AND < 10% of requirement	0.5%
≥ Rs 1 lakh OR ≥ 10% of requirement	1.0%

If the shortfall persists for more than 3 consecutive days, or occurs more than 5 times in a month, a higher rate (illustratively 5%) may apply.

WORKED

Requirement	Rs 2,19,375	
Available	Rs 1,85,000	
Shortfall	Rs 34,375	(15.7% of requirement)

→ 15.7% exceeds the 10% threshold → 1.0% rate
Penalty for one day = 34,375 x 0.01 = Rs 343.75
Plus GST on the penalty.

If it persists four days and you also get hit with the higher rate: 34,375 x 0.05 x 4 = Rs 6,875 plus GST.

NOTE THE PERVERSITY

You are most likely to face a shortfall exactly when the market has moved against you and margin requirements have risen. The penalty therefore arrives on your worst day, on top of the loss itself.

ALSO: your broker will square off the position to cure the shortfall. That square-off happens at the broker's discretion, at market, possibly at the worst price of the day, and the broker may charge a square-off fee on top.

Penalty rates illustrative. As at late 2026 - verify at nseindia.com and in your broker's tariff schedule.

Physical delivery margins in expiry week

This is where beginners get genuinely hurt

Single-stock F&O in India settles by **physical delivery**. If you hold an in-the-money stock option or any stock futures position into expiry, you take or give actual shares. In the days before expiry the exchange levies escalating **delivery margins** on positions likely to go to delivery, because the settlement obligation is for the full contract value, not the margin.

TEXT

DELIVERY MARGIN RAMP - ILLUSTRATIVE
═══════════════════════════════════

Hypothetical share at Rs 1,800, lot 500.
You are SHORT 1 lot of the 1,750 CE (in the money).
If it stays ITM, you must DELIVER 500 shares at Rs 1,750.
Settlement obligation = 1,750 x 500 = Rs 8,75,000

Delivery margin is phased in over the final sessions. An
illustrative ramp applied to the delivery obligation:

Session	% of delivery margin applied
Expiry minus 4 days	10%
Expiry minus 3 days	25%
Expiry minus 2 days	45%
Expiry minus 1 day	70%
Expiry day	100%

Expiry minus 1 day requirement, illustrative:
delivery margin base (say 20% VaR-type figure on
Rs 8,75,000) = Rs 1,75,000, x 70% = Rs 1,22,500
ON TOP OF your normal SPAN + exposure margin.

WHAT GOES WRONG

1. You sold a stock call for Rs 8,000 of premium.
2. It drifts in the money.
3. Three days before expiry your margin requirement DOUBLES because delivery margin kicks in.
4. You have no spare cash. Shortfall penalty. Forced exit.
5. Or you hold on and get assigned - and must deliver 500 shares you do not own. That means buying them in the market (or facing AUCTION SETTLEMENT at a punitive price).

AUCTION / CLOSE-OUT

If you cannot deliver, the exchange buys the shares in an auction market on your behalf and charges you the cost plus a penalty. Auction prices can be materially worse than the market price. This is not a theoretical risk.

THE SIMPLE RULE THAT AVOIDS ALL OF THIS

CLOSE ALL SINGLE-STOCK F&O POSITIONS BEFORE EXPIRY WEEK unless you specifically intend to take or give delivery, have the shares or the full cash available, and understand the margin ramp.

Delivery margin percentages and schedule illustrative.
As at late 2026 - verify at nseindia.com.

Collateral: cash versus pledged securities

Point	Detail (illustrative, verify)
You can pledge securities as margin	Shares, ETFs and mutual fund units in your demat can be pledged to the broker, who re-pledges to the clearing corporation. You receive collateral value after a haircut.
Haircut	The collateral value is less than market value — e.g. a 20% haircut on a liquid share means ₹1,00,000 of shares gives ₹80,000 of margin. Haircuts are larger for volatile or less liquid securities.
Cash component requirement	A minimum proportion of total margin — commonly cited as 50% — must be in cash or cash equivalents. You cannot fund an entire derivatives book with pledged shares.
MTM must be settled in cash	Daily futures losses are debited in cash. Pledged shares do not pay your MTM. This catches people who pledge a portfolio and then cannot fund a losing day.
Pledging is not free	Depositories and brokers levy per-instruction pledge/unpledge charges.
Double jeopardy	If you pledge equity to fund a derivatives short and the market falls, the pledged collateral's value falls AND the derivatives position loses AND the haircut may widen. Correlated collateral is dangerous collateral.

⚠ **Pledging your long-term portfolio to fund F&O**
This is extremely common in India and it is how a bad trading period becomes a destroyed investment portfolio. The moment your equity holdings are collateral for a leveraged derivatives book, they are no longer long-term holdings — they are liquidation fodder in the event of a margin call, and the margin call will arrive on the day the market is down. Verify current pledge rules and cash-component requirements at sebi.gov.in and with your broker.

The real cost of trading options

Transaction costs are not a rounding error in derivatives. In SEBI's own analysis they account for a substantial share of what small traders lose. The problem is that they are small in absolute rupees and large as a percentage of the premium you are actually risking — which is exactly the kind of number that human beings systematically ignore.

⚠ **Every rate below is illustrative and dated**

Securities Transaction Tax rates, exchange transaction charges, SEBI turnover fees, stamp duty and GST have all been revised in recent years. The figures used here are plausible for **late 2026** and are intended to teach the *structure* of the cost stack and the arithmetic of the drag. Get current rates from **nseindia.com**, **incometax.gov.in** and your broker's published tariff before computing anything you will act on.

The cost stack

Charge	Levied on	Applies to	Illustrative basis (verify)
Brokerage	Per executed order	Both sides	Flat fee per order (discount brokers) or a percentage; a flat per-order fee plus GST is the common Indian model
STT on options — sale	**Premium** value	Seller of the option	A small percentage of premium turnover on the sell side
STT on options — **exercise**	**Intrinsic/settlement** value, not premium	Buyer, on an option that expires in the money	A percentage of the *settlement* value — this is the trap, worked below
STT on futures	Traded value	Seller	A small percentage of the sell-side turnover
Exchange transaction charges	Premium turnover (options) / traded value (futures)	Both sides	Set by the exchange; revised periodically; options charged on premium
SEBI turnover fee	Turnover	Both sides	A very small per-crore levy
Stamp duty	Buy side only	Buyer	A small per-crore rate, uniform across states since the 2020 unification
GST	Brokerage + exchange charges + SEBI fees	Both sides	18% on the service components
Bid–ask spread	Every fill	Both sides	Not a "charge" but a real cost — often the largest single item
Slippage	Every fill	Both sides	The difference between the price you saw and the price you got

Worked round-trip cost on a typical retail position

TEXT

```
ROUND-TRIP COST: BUY AND SELL ONE LOT OF AN INDEX OPTION
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Position: BUY 1 lot 25,300 CE @ 96.25, sell it later @ 120.00
Lot size 75.

TURNOVER
  Buy premium turnover  =  96.25 x 75 = Rs  7,218.75
  Sell premium turnover = 120.00 x 75 = Rs  9,000.00
  Total premium turnover              = Rs 16,218.75

COST LINE BY LINE (illustrative rates - VERIFY)
─────────────────────────────────────────────────

1. Brokerage
   Flat Rs 20 per executed order, 2 orders        Rs  40.00

2. STT (options sale side, illustrative 0.1% of
   premium on sell)
   9,000.00 x 0.001                               Rs   9.00

3. Exchange transaction charges
   (illustrative 0.035% of premium turnover)
   16,218.75 x 0.00035                            Rs   5.68

4. SEBI turnover fee
   (illustrative Rs 10 per crore of turnover)
   16,218.75 x 0.000001                           Rs   0.02

5. Stamp duty (buy side, illustrative Rs 300/crore
   = 0.003%)
   7,218.75 x 0.00003                             Rs   0.22

6. GST @ 18% on (brokerage + exchange chg + SEBI fee)
   (40.00 + 5.68 + 0.02) x 0.18                   Rs   8.23
                                                  ──────────
   EXPLICIT CHARGES SUBTOTAL                      Rs  63.15

7. BID-ASK SPREAD - the item nobody counts
   Suppose the 25,300 CE is quoted 95.50 / 97.00.
   Mid = 96.25. You buy at the ask (97.00) and
   later sell at the bid.
   Spread cost per round trip = 1.50 points
   1.50 x 75                                      Rs 112.50
                                                  ──────────
   TRUE ROUND-TRIP COST                           Rs 175.65

NOW THE PERCENTAGE THAT MATTERS
─────────────────────────────────

Capital at risk (premium paid)       Rs 7,218.75
```

```
True round-trip cost                 Rs   175.65
COST AS % OF CAPITAL AT RISK              2.43%

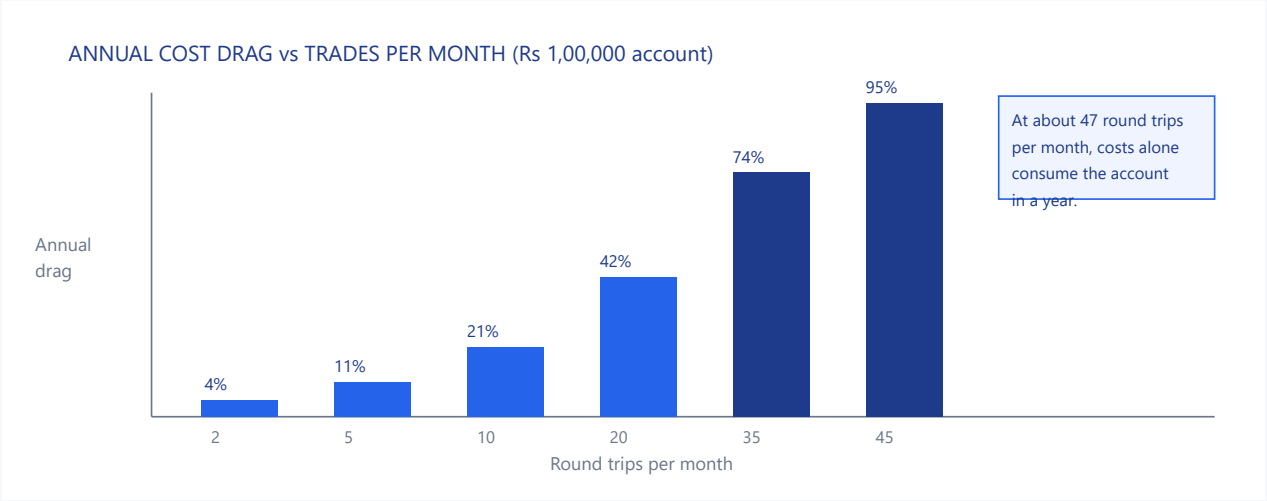
You must make 2.43% on the premium just to break even.
In index points, the option must rise by
  175.65 / 75 = 2.34 points before you see a rupee.

NOW DO IT TWENTY TIMES A MONTH
_____________________________

20 round trips x Rs 175.65 = Rs 3,513 per month
                           = Rs 42,156 per year

On a Rs 1,00,000 trading account that is a 42% ANNUAL DRAG.
Your strategy must generate +42% a year gross simply to
break even net.

Now consider that you are trading a near-zero-sum game against
professionals, and the magnitude of the 90% loss statistic
stops being mysterious.
```



Transaction cost drag scales linearly with trade frequency. At retail option sizes, two or three trades a day is an account-destroying cost structure before any market risk is taken. Rates illustrative.

The STT-on-exercise trap

This one has wiped out profitable positions

Securities Transaction Tax on a **sold** option is charged on the **premium**. STT on an option that is **exercised** (i.e. expires in the money and settles) is charged on the **settlement / intrinsic value**, which can be orders of magnitude larger than the premium. If you let a deep in-the-money option expire rather than selling it, you can hand over a large STT bill on a position you thought was a small win.

TEXT

```
STT ON EXERCISE - THE ARITHMETIC
================================
Illustrative rate: STT on exercise of an option = 0.125% of the
INTRINSIC (settlement) value. VERIFY the current rate and base
at incometax.gov.in / nseindia.com - this rate has been revised.

SCENARIO: you bought a deep ITM call and let it expire.
  BUY 1 lot 24,000 CE @ 980.00, lot size 75
  Cost = 980 x 75 = Rs 73,500

  Index settles at 25,000 on expiry.
  Intrinsic value = 25,000 - 24,000 = 1,000 points
  Settlement value = 1,000 x 75 = Rs 75,000

ROUTE A - YOU SELL THE OPTION BEFORE EXPIRY at 1,000.00
  Proceeds           = 1,000 x 75 = Rs 75,000
  STT on sale (illustrative 0.1% of premium)
                     = 75,000 x 0.001 = Rs 75
  Gross P&L          = 75,000 - 73,500 = Rs 1,500
  Net after STT      = Rs 1,425

ROUTE B - YOU LET IT EXPIRE AND IT IS EXERCISED
  Settlement receipt = Rs 75,000
  STT on exercise    = 75,000 x 0.00125 = Rs 937.50
  Gross P&L          = Rs 1,500
  Net after STT      = Rs 562.50

  You lost 62% of your profit by doing nothing.

NOW MAKE IT WORSE - A LARGER ITM POSITION
  BUY 1 lot 23,000 CE @ 1,990.00 → cost Rs 1,49,250
  Index settles at 25,000. Intrinsic = 2,000 points
  Settlement value = Rs 1,50,000
  Gross P&L = Rs 750

  Route A (sell at 2,000): STT = 150,000 x 0.001 = Rs 150
    Net = Rs 600 PROFIT

  Route B (let it expire): STT = 150,000 x 0.00125 = Rs 1,875
    Net = 750 - 1,875 = Rs 1,125 LOSS

  A PROFITABLE POSITION BECAME A LOSS BECAUSE OF SETTLEMENT
  MECHANICS, NOT BECAUSE OF THE MARKET.

THE RULES THAT FOLLOW
  1. SELL your in-the-money options in the market before expiry
     rather than letting them be exercised, unless you have
     computed that exercise is cheaper.
```

```
  2. The deeper in the money, the worse the penalty, because
     STT on exercise scales with INTRINSIC VALUE while your
     PROFIT does not.
  3. Check whether your broker offers a "do not exercise" (DNE)
     instruction. Availability and applicability have changed
     over time - at one point DNE was permitted for CTM
     (close-to-the-money) options specifically. VERIFY current
     availability with your broker and at nseindia.com.
  4. Set a calendar reminder for expiry day on every open
     option position. "I forgot" is an expensive sentence.

STT rates illustrative. As at late 2026 - verify current rates
and bases at incometax.gov.in and nseindia.com.
```

Cost as a function of position size

TEXT

```
WHY SMALL ACCOUNTS ARE PUNISHED HARDEST
═══════════════════════════════════════

Flat per-order brokerage (say Rs 20 + GST = Rs 23.60) is the
same whether you trade 1 lot or 20 lots. The bid-ask spread
scales with size but the spread as a PERCENTAGE is roughly
constant. So:

Premium    Lots  Prem.turnover  Flat broker  Spread   Total   % of
per unit         (buy, Rs)      (2 orders)   cost     cost    risk
────────  ────  ─────────────  ──────────  ───────  ──────  ─────
   15.00     1        1,125.00       47.20   112.50   163.0  14.5%
   96.25     1        7,218.75       47.20   112.50   166.0   2.3%
   96.25     5       36,093.75       47.20   562.50   630.0   1.7%
   96.25    20     1,44,375.00       47.20  2,250.00 2,320.0   1.6%
  218.50     1       16,387.50       47.20   112.50   166.0   1.0%

READ THE FIRST ROW. Buying a Rs 15 "cheap" option costs you
14.5% of your capital in round-trip friction. The option must
rise 14.5% before you make a single rupee. On an option with
a 13% probability of finishing in the money.

THE CHEAPER THE OPTION, THE WORSE THE COST RATIO. This is the
exact opposite of what "cheap options are low risk" suggests,
and it is why expiry-day scalping of far-OTM options is a
near-certain way to lose money over a sample of trades.

Illustrative. Spread assumptions vary by strike liquidity.
```

Bid–ask spread: the cost you cannot see on a contract note

Contract type	Typical quoted spread (illustrative)	Spread as % of premium	Implication
ATM index option, near expiry, liquid index	0.25–1.00 points	0.2–0.5%	Tradeable; the only part of the chain where frequent trading is even arithmetically conceivable
2–3% OTM index option	0.50–2.00 points	1–4%	Meaningful drag; avoid frequent round trips
Far OTM index option (₹5–20 premium)	0.25–1.00 points	5–20%	Prohibitive. You lose a fifth of your stake on entry and exit.
Single-stock options, liquid name	Wider than index	2–6%	Spreads and multi-leg structures become expensive fast
Single-stock options, less liquid name	Very wide, sometimes no two-way quote	10%+	You may be unable to exit at any sensible price. Avoid.
Far-month options, any underlying	Wide	5%+	Calendar spreads carry hidden exit cost on the back leg

✦ **A discipline that costs nothing**

Always use limit orders, never market orders, in options. A market order in a wide-spread strike is an invitation to be filled at the worst available price. If you cannot get filled at your limit, that is information: the liquidity you assumed does not exist, and you should not be in that contract.

Assignment, expiry and settlement

More retail damage is done in the last three days of a contract's life than in the preceding three weeks. This section is the operational manual for the end of the cycle.

What happens at expiry, step by step

1 **Trading stops**

Trading in the expiring contract ceases at the end of the expiry-day session. There is no extended window.

2 **Settlement price is determined**

For index derivatives, typically the volume-weighted average price of the underlying index over a defined closing period of the expiry day — not the last traded price. For single stocks, the closing price of the underlying.

3 **Out-of-the-money options expire worthless**

Nothing happens. The premium paid is lost in full by the buyer and retained in full by the seller. No STT on exercise, no settlement.

4 **In-the-money options are automatically exercised**

You do not need to do anything. Indian exchanges automatically exercise all ITM options at expiry (subject to any applicable do-not-exercise facility). This is why "I did not want to exercise" is not a defence.

5 **Index options settle in cash**

The intrinsic value is credited to the buyer and debited from the seller. STT on exercise applies to the buyer on the settlement value.

6 **Stock options settle by physical delivery**

The call buyer receives shares and pays the strike; the call seller delivers shares and receives the strike. The put buyer delivers shares and receives the strike; the put seller receives shares and pays the strike.

7 **Failure to deliver triggers auction**

If you are obliged to deliver shares you do not hold, the exchange procures them in the auction market at your cost, plus penalties.

TEXT

```
INDEX OPTION CASH SETTLEMENT - WORKED
═══════════════════════════════════════

You are LONG 1 lot 24,800 CE. Lot size 75.
Settlement value of the index on expiry (VWAP-based): 25,062.40

Intrinsic value = 25,062.40 - 24,800 = 262.40 points
Cash credited   = 262.40 x 75 = Rs 19,680.00

STT on exercise (illustrative 0.125% of settlement value):
  Settlement value for STT purposes = 19,680.00
  STT = 19,680 x 0.00125 = Rs 24.60

Net credit = Rs 19,655.40

If you had paid 310.00 for this option (Rs 23,250), your
net result is 19,655.40 - 23,250 = -Rs 3,594.60.
IN THE MONEY AND STILL A LOSS - because you paid 310 for
something worth 262.40 at expiry. Being ITM is not the same
as being profitable. Your break-even was 25,110.

A NOTE ON THE SETTLEMENT PRICE
  It is a VWAP over a closing window, not the last print. On a
  volatile expiry afternoon the VWAP can differ from the close
  by tens of points. You cannot trade against the settlement
  price because it is computed after trading ends.

As at late 2026 - verify the settlement price methodology and
STT rate at nseindia.com and incometax.gov.in.
```

TEXT

```
PHYSICAL SETTLEMENT OF STOCK OPTIONS - FOUR CASES
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Hypothetical share, lot size 500, settles at Rs 1,850 on expiry.

CASE 1: You are LONG a 1,800 CE (ITM by 50)
  You must PAY  1,800 x 500 = Rs 9,00,000
  You RECEIVE   500 shares (worth Rs 9,25,000)
  Net benefit   Rs 25,000 less STT on exercise
  REQUIREMENT: Rs 9,00,000 IN CASH. Not margin. Cash.
  If you do not have it, your broker will square you off before
  expiry or you face a funds shortfall and penalties.

CASE 2: You are SHORT a 1,800 CE (ITM by 50)
  You must DELIVER 500 shares
  You RECEIVE      Rs 9,00,000
  REQUIREMENT: 500 shares in your demat, unencumbered.
  If you do not have them: short delivery → auction → you pay
  the auction price plus penalty, which can far exceed Rs 1,850.

CASE 3: You are LONG a 1,900 PE (ITM by 50)
  You must DELIVER 500 shares
  You RECEIVE      1,900 x 500 = Rs 9,50,000
  REQUIREMENT: 500 shares in demat. Same auction risk as Case 2.

CASE 4: You are SHORT a 1,900 PE (ITM by 50)
  You must PAY   Rs 9,50,000
  You RECEIVE    500 shares
  REQUIREMENT: Rs 9,50,000 in cash.

THE PATTERN
  TWO of the four cases require you to produce LARGE CASH.
  TWO of the four require you to produce SHARES YOU MAY NOT OWN.
  None of the four is satisfied by the margin you posted.

A position with Rs 40,000 of margin can create a Rs 9.5 lakh
settlement obligation. That ratio is the whole hazard.

THE OPERATIONAL RULE
  Put a recurring calendar entry three trading days before every
  monthly expiry. Review every single-stock F&O position. Close
  anything you do not intend to settle physically. Do not rely
  on your broker's auto square-off, which happens at the
  broker's chosen time and price and may carry a fee.
```

Do-not-exercise (DNE)

At various points Indian exchanges have permitted a "do not exercise" instruction for close-to-the-money options, so that a buyer whose option is barely in the money can avoid a settlement whose STT and delivery costs exceed the intrinsic value. The availability, the eligible strikes and the broker-level support for this have changed over time. Check with your broker and at nseindia.com whether DNE is currently available and for which contracts. Do not assume it exists.

Expiry-day mechanics to understand before you trade one

Phenomenon	What happens	Why it matters
Theta collapse	All remaining time value must reach zero by the close	Option buyers face a decay rate that is effectively vertical in the final hours
Gamma explosion	Near-ATM deltas swing between 0 and 1 on small moves	Short positions can change character completely in minutes; see the gamma worked example
Pinning	Price often gravitates toward strikes with the largest open interest	Partly a hedging artefact from market makers rebalancing. Not reliable enough to trade, and widely over-claimed.
Spread widening	Market makers widen quotes as uncertainty concentrates	Your exit costs more exactly when you most need to exit
Additional tail-risk margin	SEBI's framework added an extra margin on short options on expiry day	Short-option expiry-day strategies require materially more capital than before. Verify current level.
Loss of calendar-spread margin benefit	Removed on expiry day	Spread strategies held into expiry now consume full margin on both legs
Settlement VWAP	Index settles on a closing-window average, not the last trade	A late spike or dip may not be reflected in your settlement

Expiry day is the professionals' home ground

Every structural feature of expiry day — maximum gamma, maximum theta, maximum spread, maximum order flow — favours a participant with automated hedging and sub-second execution. It is the single worst arena for a discretionary retail trader using a mobile app, and SEBI's data shows it is precisely where the largest concentration of retail losses occurs.

Why most retail F&O traders lose: the arithmetic

The loss rate is not a mystery and it is not primarily about psychology, though psychology makes it worse. It is a stack of independently sufficient mathematical reasons, any one of which would produce a negative expectancy on its own. Together they produce a near-certainty.

The eight reasons, in order of magnitude

#	Reason	Effect on expectancy
1	**Negative sum after costs**	The game pays out less than it takes in. Average participant loses by the amount of the friction, before any skill differential.
2	**Adverse counterparty selection**	You are trading against the participants who price these instruments for a living. The average retail fill is worse than the average professional fill.
3	**Leverage mis-sizing**	Position sizes are chosen by premium outlay, not by delta-equivalent exposure, producing 3–5x the intended risk.
4	**Selling tails**	High-win-rate, unbounded-loss structures create a long run of small wins terminated by one catastrophic loss.
5	**Over-trading**	Cost drag scales linearly with frequency; a 40%+ annual drag is routine at retail sizes.
6	**Behavioural errors**	Revenge trading, averaging down, cutting winners early, holding losers, position-size escalation after losses.
7	**Survivorship bias in the information diet**	You only see the winning screenshots. Your estimate of achievable returns is calibrated on a non-random sample.
8	**Instrument complexity**	Most participants do not know their position's vega or gamma, and therefore do not know what they own.

1–2. The expectancy equation, stated properly

TEXT

THE EXPECTANCY EQUATION FOR ANY TRADING ACTIVITY

$$E = (P_{win} \times Avg_{win}) - (P_{loss} \times Avg_{loss}) - Costs$$

For a PURE zero-sum game with no skill differential:

$(P_{win} \times Avg_{win}) = (P_{loss} \times Avg_{loss})$

Therefore $E = -Costs$, always negative.

To have $E > 0$ you need:

$(P_{win} \times Avg_{win}) - (P_{loss} \times Avg_{loss}) > Costs$

i.e. your skill edge must EXCEED the friction. Not match it. Exceed it, persistently, across hundreds of trades, net of the fact that your counterparty is also trying to have an edge.

QUANTIFYING THE HURDLE FOR A TYPICAL RETAIL OPTION TRADER

Account	Rs 1,00,000
Average position premium	Rs 7,000
Round trips per month	20
Round-trip cost per trade	Rs 176 (computed earlier)
Monthly cost	Rs 3,520
Annual cost	Rs 42,240 = 42.2% of account

For $E > 0$ you need a gross edge of more than 42% a year on account equity, or 2.5% per trade on premium, purely to overcome friction.

WHAT 2.5% PER TRADE MEANS

If your average winner is +30% of premium and your average loser is −40% of premium (typical for directional option buying), then to net +2.5% you need:

$$0.30\,w - 0.40\,(1-w) = 0.025$$
$$0.30\,w - 0.40 + 0.40\,w = 0.025$$
$$0.70\,w = 0.425$$
$$w = 0.607$$

You need a 61% WIN RATE on directional option trades, every month, forever, just to break even after costs.

For context: the implied probability that an ATM option finishes in the money is roughly 50%. You are being asked to beat a coin flip by 11 percentage points, persistently, against professionals.

THAT IS THE ENTIRE GAME, EXPRESSED IN ONE NUMBER.

3. Position sizing: the error almost everyone makes

TEXT

```
SIZING BY PREMIUM vs SIZING BY EXPOSURE
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Account: Rs 2,00,000. You decide "I will risk 10% per trade,
so Rs 20,000 of premium."

TRADE: buy 25,600 CE @ 34.50 (delta 0.13), lot size 75
  Premium per lot = 34.50 x 75 = Rs 2,587.50
  Rs 20,000 / 2,587.50 = 7.7 → you buy 7 lots
  Premium outlay = Rs 18,112.50  ("only 9% of the account")

WHAT YOU ACTUALLY OWN
  Delta-equivalent units = 7 lots x 75 x 0.13 = 68.25 units
  Notional exposure      = 68.25 x 25,000 = Rs 17,06,250

  You have Rs 17 lakh of index exposure on a Rs 2 lakh account.
  That is 8.5x leverage, achieved while telling yourself you
  risked 9%.

NOW ADD GAMMA
  Gamma per unit 0.00021. If the index rallies 400 points:
  New delta = 0.13 + (0.00021 x 400) = 0.214
  Delta-equivalent = 7 x 75 x 0.214 = 112.4 units
  Notional = Rs 28,09,000

  Your exposure grew 65% without you doing anything.

AND THE DOWNSIDE
  If the index finishes below 25,600, you lose Rs 18,112.50 -
  100% of the premium, 9% of the account - on a trade that had
  roughly a 13% probability of finishing in the money.

  Expected value, crudely:
    0.13 x (some average ITM payoff, say Rs 30,000)
      = +Rs 3,900
    0.87 x (-Rs 18,112.50)
      = -Rs 15,758
    EV = -Rs 11,858 per trade, before costs.

  Repeat that eight times and the account is gone.

THE CORRECT SIZING QUESTION
  Not "how much premium am I spending?"
  But  "how many delta-equivalent units of the index do I own,
        and what happens to my account if the index moves 3%?"
```

4. The maths of "sell options for steady income" — a full blow-up scenario

TEXT

```
A 24-CYCLE SIMULATION OF SHORT STRANGLE "INCOME"
================================================
Starting capital                Rs 5,00,000
Strategy: sell a ~2.5% OTM strangle each cycle
Margin per strangle (illustrative) Rs 2,40,000
Position: 1 strangle per cycle (conservative - most people
          scale up after a winning run, which is the problem)
Premium collected per cycle     Rs 6,000 (net, after costs)

CYCLES 1-14: THE MARKET BEHAVES
  14 x Rs 6,000 = +Rs 84,000
  Capital now Rs 5,84,000
  Return so far: +16.8%
  Win rate so far: 100%

  WHAT HAPPENS PSYCHOLOGICALLY HERE
  - You conclude the strategy works.
  - You start posting your results.
  - You scale from 1 strangle to 3 ("I have more capital now
    and I have proven the edge").
  - Margin for 3 strangles = Rs 7,20,000 - more than you have,
    so you PLEDGE your equity portfolio to fund it.
  - Premium per cycle now Rs 18,000.

CYCLES 15-20: STILL FINE
  6 x Rs 18,000 = +Rs 1,08,000
  Capital now Rs 6,92,000
  Return: +38.4% over 20 cycles. Win rate 100%.
  You are now certain. You tell friends.

CYCLE 21: A 6% GAP DOWN OVERNIGHT
  Index gaps from 25,000 to 23,500.
  Your short 24,400 PE (sold at 46.00) is now worth
  roughly 950.00 intrinsic plus elevated time value, say 1,080.

  Loss per put = (1,080 - 46) x 75 = Rs 77,550
  You are short 3 of them       = Rs 2,32,650

  Your short calls expire worthless: +3 x 34.50 x 75 = +Rs 7,763
  NET LOSS ON THE CYCLE         = -Rs 2,24,887

  AND SIMULTANEOUSLY:
  - IV spikes from 14% to 30%, so even if you want to close,
    the exit price is far worse than intrinsic alone.
  - SPAN margin requirement roughly doubles.
  - Your PLEDGED EQUITY COLLATERAL has also fallen 6%, and the
    haircut may have widened, so your available margin falls.
  - You face a margin shortfall, a penalty, and a forced
```

```
    square-off at the worst available price.

AFTER CYCLE 21
  Capital: 6,92,000 - 2,24,887 = Rs 4,67,113
  You are now BELOW your Rs 5,00,000 starting capital after
  20 consecutive wins and one loss.

  Win rate: 20/21 = 95.2%
  Return: -6.6%

CYCLE 22: THE AFTERMATH
  Markets remain volatile. You either:
  (a) stop, having learned an expensive lesson, or
  (b) "make it back" by increasing size further - which is how
      the -6.6% becomes -60%.

Option (b) is what the SEBI data is measuring in aggregate.

THE GENERAL RESULT
  With a strategy whose max gain is G and whose tail loss is
  L, and whose tail probability is q, the strategy is
  profitable only if:

      (1-q) x G  >  q x L

  With G = Rs 18,000, L = Rs 2,25,000, that requires:
      (1-q) x 18,000 > q x 225,000
      18,000 - 18,000q > 225,000q
      18,000 > 243,000q
      q < 0.074

  The tail event must occur less than 7.4% of the time - i.e.
  less often than roughly once in 14 cycles. A 6% adverse gap
  in an Indian index over a monthly cycle is not a once-in-14
  event in a long sample. It is closer to once in 10-20, and
  the events cluster.

Figures illustrative throughout. The SHAPE of this result is
robust and is the arithmetic behind SEBI's loss statistics.
```

The central deception of high-win-rate strategies

A 95% win rate feels like skill. It is a property of the *structure*, not of the trader. Anyone selling far-out-of-the-money options achieves a high win rate automatically. The win rate tells you nothing about expectancy because it says nothing about the size of the loss. When someone shows you a track record, ask for the maximum drawdown and the largest single loss, not the win rate.

5–6. Over-trading, revenge trading and averaging down

TEXT

```
THE MARTINGALE / AVERAGING-DOWN TRAP ON OPTIONS
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"The option fell from 100 to 50 – I will double up and my
average becomes 75, so I only need it back to 75."

WHY THIS IS CATASTROPHIC ON OPTIONS SPECIFICALLY
  1. An option has an EXPIRY. A share can recover over years;
     an option has days. Time is running out while you average.
  2. The fall from 100 to 50 was caused by some combination of
     delta (wrong direction), theta (time passing – which
     continues) and vega (IV falling – often permanently after
     an event). Two of those three are IRREVERSIBLE.
  3. Your position size has doubled while your probability of
     success has fallen.

WORKED SEQUENCE
  Buy 1 lot @ 100.00    outlay Rs  7,500   total Rs  7,500
  Falls to 50: buy 2    outlay Rs  7,500   total Rs 15,000
                        (3 lots, avg 66.67)
  Falls to 25: buy 4    outlay Rs  7,500   total Rs 22,500
                        (7 lots, avg 42.86)
  Falls to 10: buy 8    outlay Rs  6,000   total Rs 28,500
                        (15 lots, avg 25.33)
  Expires at 0:         LOSS Rs 28,500

  You started risking Rs 7,500 and lost Rs 28,500 – 3.8x your
  intended risk – on a single idea that was wrong from the start.

  To have broken even on the final position, the option needed
  to reach 25.33 from 10.00 – a 153% rise – with less time left
  than when you first bought it at 100.

THE STRUCTURAL POINT
  Averaging down converts a bounded loss into a much larger
  bounded loss and converts a small mistake into an account
  event. On options, where the instrument expires, it has no
  defensible version.

COMPARE: averaging down on a diversified index fund during a
drawdown is a reasonable long-horizon behaviour. The two
activities share a name and nothing else.
```

Behaviour	What triggers it	The cost, quantified
Revenge trading	A loss that felt unfair; a stop hit just before the market turned	Position size typically 2–3x normal, entered without the usual criteria. One revenge trade can exceed a month of careful gains.
Over-trading from boredom	A quiet session; the need to "do something"	Each unnecessary round trip costs ₹176 at retail size. Ten a month from boredom = ₹21,000 a year.
Cutting winners early	Fear of giving back a gain; the relief of booking a profit	Truncates the right tail. Option strategies depend on a few large winners; cutting at +20% destroys the distribution.
Holding losers	Loss aversion; "it will come back"	On options, "it will come back" has a deadline. Holding converts a −40% into a −100%.
Size escalation after a win streak	Overconfidence; mistaking a structural win rate for skill	Guarantees that your largest position is on at the moment of your worst outcome. See the blow-up scenario.
Expiry-day "lottery" trades	Cheap premium; the feeling that a small stake is low risk	Cost ratio 10–20% of stake, win probability often under 15%, EV deeply negative. Repeated weekly, this is the fastest route to zero.

7. Survivorship bias in your information diet

⚠ **What you see is not a random sample**

If 100 people trade a weekly expiry and 7 make money, you will see screenshots from the 7. The 93 post nothing. Social platforms amplify extreme positive outcomes because those generate engagement. Your intuition about "what is achievable" is being calibrated on the visible tail of a distribution whose body is invisible to you. This is not a minor distortion — it is the single largest source of unrealistic expectations in retail trading.

- ☐ Screenshots are trivially fabricated, and even genuine ones show one position, not an account.
- ☐ A genuine ₹3 lakh profit screenshot is compatible with a ₹9 lakh year-to-date loss.
- ☐ Nobody posts the margin call. Nobody posts the pledged portfolio that got liquidated.

- ☐ Paid "mentors" earn from subscriptions, not from trading. Their incentive is your enrolment, not your P&L.
- ☐ The only track record worth anything is a full, audited, continuous equity curve with drawdowns — which essentially nobody in retail social media provides.

8. Not knowing what you own

> ℹ **A simple diagnostic**
> Open your current F&O positions. Without looking anything up, write down: net delta in index units, net theta in rupees per day, net vega in rupees per IV point, and the rupee loss if the index gaps 4% against you overnight. If you cannot produce those four numbers in under a minute, you do not know what you own. That is not a moral failing — it is a reason to stop trading the instrument until you can.

The SEBI regulatory response, 2024–25

In October 2024 SEBI announced a package of measures to "strengthen the equity index derivatives framework for increased investor protection and market stability", phased in from November 2024 through 2025. It is the most significant intervention in Indian retail derivatives in a decade and it exists because of the loss data in section one.

> ⚠ **Verify the current framework**
> The measures below are described as announced and as subsequently implemented, **as at late 2026**. Several were phased, some thresholds have been revised after implementation, and further consultation papers have been issued. The authoritative source is **sebi.gov.in** (press releases, circulars and consultation papers) and the exchange circulars at **nseindia.com**. Do not rely on this summary for a trading decision.

The six measures

#	Measure	What changed	Stated rationale
1	**Upfront collection of option premium from buyers**	Brokers must collect the full option premium from the buyer upfront, removing intraday leverage on option purchases	Prevent undue intraday leverage and ensure no position is taken without adequate funds
2	**Removal of calendar-spread margin benefit on expiry day**	Positions expiring on a given day no longer receive offsetting margin treatment against other expiries on that day	The basis risk between an expiring and a non-expiring leg is real and was being under-margined
3	**Intraday monitoring of position limits**	Exchanges monitor client-level position limits intraday via random snapshots, not just at end of day	Prevent participants exceeding limits intraday and unwinding before measurement
4	**Minimum contract size raised**	The minimum notional value of a derivatives contract was increased substantially, with lot sizes revised upward accordingly; a further phased increase was indicated	Ensure participants taking leveraged positions have commensurate risk-taking ability; price smaller retail out of the product
5	**Rationalisation of weekly index derivatives**	Weekly options expiry limited to **one benchmark index per exchange**, ending the regime of near-daily expiries across multiple indices	Reduce the volume of very short-dated, speculative, high-gamma activity
6	**Additional tail-risk margin on expiry day**	An extra margin levied on short option positions on the day of expiry	Expiry-day short-option risk is materially higher than standard margin models capture

What each measure actually does to a retail position

TEXT

```
THE MEASURES, TRANSLATED INTO EFFECTS ON YOUR ACCOUNT
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1. UPFRONT PREMIUM COLLECTION
   BEFORE: some brokers allowed buying options against
           unsettled funds or notional limits intraday.
   AFTER : you need the cash in the account. Full stop.
   EFFECT: removes a source of uncontrolled intraday leverage.
           Mildly inconvenient; unambiguously protective.

2. CALENDAR-SPREAD MARGIN BENEFIT REMOVED ON EXPIRY DAY
   BEFORE: a near/far spread got substantial margin relief
           even on expiry day.
   AFTER : full margin on both legs that day.
   EFFECT: capital required for spread strategies held into
           expiry roughly doubles. Many such strategies became
           uneconomic for small accounts.

3. INTRADAY POSITION LIMIT MONITORING
   BEFORE: limits measured end of day; intraday excesses
           invisible.
   AFTER : random intraday snapshots; breaches penalised.
   EFFECT: mostly affects larger participants; for retail it
           closes a loophole you probably were not using.

4. MINIMUM CONTRACT SIZE RAISED
   BEFORE: a single index option lot might have had a notional
           of roughly Rs 5 lakh, with premiums of a few
           thousand rupees.
   AFTER : minimum notional raised substantially; one lot now
           represents a much larger exposure.
   EFFECT: this is the BIGGEST change for small accounts.
           - A single lot now costs more in premium
           - Position sizing in anything under 1 lot is
             impossible
           - Hedging a small portfolio became harder, not
             easier (the granularity problem)
           - Small accounts were, in effect, priced out
   HONEST ASSESSMENT: protective in intent, but it also made
           legitimate small-scale hedging harder. SEBI's
           position is that if you cannot afford one contract,
           you cannot afford the risk.

5. WEEKLY EXPIRY RATIONALISATION
   BEFORE: multiple indices with weekly expiries across two
           exchanges produced an expiry nearly every weekday.
   AFTER : one benchmark index per exchange has weekly options.
   EFFECT: the "every day is expiry day" scalping regime ended.
           Volumes in ultra-short-dated options fell materially.
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            This is the measure most directly aimed at the
            behaviour the loss data identified.

6. ADDITIONAL EXPIRY-DAY TAIL-RISK MARGIN
   BEFORE: standard SPAN margin on expiry day.
   AFTER : extra margin on SHORT option positions that day.
   EFFECT: expiry-day option selling requires materially more
           capital. Directly targets the gamma-risk problem
           described earlier in this guide.

As at late 2026 - verify current implementation at sebi.gov.in.
```

Did it work?

The honest answer is partially, and the evidence is still being gathered.

WHAT APPEARS TO HAVE CHANGED

Index option volumes in the very short-dated segment fell after the weekly-expiry rationalisation. The number of unique individual participants in index options declined from the peak. Average contract size rose, mechanically, from the minimum-size increase.

WHAT DID NOT CHANGE

SEBI's own 2025 follow-up study covering FY 2024–25 found roughly 91% of individual index-option traders still losing money. The loss rate barely moved. Fewer participants, similar proportion losing.

WHY THE LOSS RATE IS STICKY

The measures reduced *access* and *leverage*. They did not change the zero-sum structure, the cost stack, or the skill gap between retail and professional participants. A smaller number of people losing at the same rate is a partial success at best.

THE DISPLACEMENT QUESTION

Some activity may have migrated to other products, other segments, or to unregulated venues (dabba trading — illegal off-exchange betting on prices, which carries total counterparty risk and no legal recourse). Regulatory tightening in a legal market does not reduce the appetite that drove the activity.

What further change might look like

- ☐ Further increases in minimum contract size, phased.
- ☐ Product suitability assessments or categorisation of derivative products by complexity.
- ☐ Revisions to expiry-day margining and position limits.
- ☐ Stricter rules on the marketing of derivatives, finfluencer regulation, and the advertising of returns.
- ☐ Possible changes to the taxation of derivative transactions (STT rates on F&O have been revised before).
- ☐ Continued publication of participant-level outcome studies, which is itself a policy instrument — disclosure as deterrent.

ℹ **Risk disclosures on broker platforms**
SEBI has required brokers and certain intermediaries to display prominent disclosures about the proportion of individual traders losing money in equity derivatives. If your broker shows you such a notice, it is not boilerplate — it is the regulator making you read the base rate before you trade. Verify the current disclosure requirement at sebi.gov.in.

Legitimate use cases, worked end to end

Four scenarios where a derivative is the right tool, worked with full numbers including costs, margin and the outcome distribution. Note that in every one of them the derivative is attached to an *existing* position or an *existing* obligation. None of them is a standalone bet.

Use case 1: hedging a ₹10 lakh equity portfolio through a defined window

TEXT

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SITUATION
═════════
Portfolio value                 Rs 10,00,000
Composition                     diversified large-cap equity
Beta vs the index               1.05
Index spot                      25,000
Lot size                        75
Horizon                         the next 24 days (a results
                                season plus a policy meeting)
Reason for hedging              you need the capital intact in
                                four months for a specific
                                real-world purpose and cannot
                                absorb a 20% drawdown now

STEP 1 - CONSIDER THE ALTERNATIVES FIRST
────────────────────────────────────────
(a) Sell 30-40% of the portfolio and hold cash.
    Cost: capital gains tax now, re-entry risk, brokerage.
    Benefit: simple, no margin, no expiry, no basis risk.
(b) Short index futures.
    Beta-adjusted exposure = 10,00,000 x 1.05 = Rs 10,50,000
    Notional per lot = 25,112 x 75 = Rs 18,83,400
    Lots needed = 0.56 → you must short 0 or 1.
    Shorting 1 lot over-hedges by 79%. REJECTED.
(c) Buy an index put.
    Finer control over the payoff; known maximum cost;
    no MTM cash calls; no margin.

(a) is often the right answer and is underrated. We work (c).

STEP 2 - CHOOSE THE STRIKE
──────────────────────────
How much are you willing to lose before protection starts?
Say 4%: the index would be at 24,000.

Buy 1 lot 24,000 PE, 24 days to expiry, IV 15.5% (skew)
Illustrative premium: 118.00 points
Cost = 118.00 x 75 = Rs 8,850
Plus charges (2 orders, stamp duty, GST): roughly Rs 60
TOTAL COST = Rs 8,910 = 0.89% of the portfolio for 24 days

STEP 3 - CHECK THE NOTIONAL MATCH
─────────────────────────────────
Put notional = 24,000 x 75 = Rs 18,00,000
Portfolio beta-adjusted    = Rs 10,50,000
Hedge ratio = 18,00,000 / 10,50,000 = 1.71x
You are OVER-HEDGED by 71% below 24,000.
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Can you fix it? Not with whole lots at this size.
Options to reduce the over-hedge:
  - accept it (the over-hedge only bites below 24,000, where
    you would otherwise be in pain anyway)
  - buy a put SPREAD instead, which caps the over-hedge benefit
    and cuts the cost
We will work both.

STEP 4A - OUTCOME TABLE, SIMPLE LONG PUT
________________________________________

Index   Index   Portfolio    Put payoff   Put cost   Net
level   move    P&L (Rs)     (Rs)         (Rs)       (Rs)
______  ______  ___________  ___________  _________  ___________
27,000  +8.0%     +84,000             0     -8,910      +75,090
26,000  +4.0%     +42,000             0     -8,910      +33,090
25,000   0.0%           0             0     -8,910       -8,910
24,500  -2.0%     -21,000             0     -8,910      -29,910
24,000  -4.0%     -42,000             0     -8,910      -50,910  <--worst
23,500  -6.0%     -63,000      +37,500      -8,910      -34,410
23,000  -8.0%     -84,000      +75,000      -8,910      -17,910
22,000 -12.0%   -1,26,000   +1,50,000      -8,910      +15,090
20,000 -20.0%   -2,10,000   +3,00,000      -8,910      +81,090

READ IT
  - Worst case is -Rs 50,910 (-5.1%) at exactly 24,000.
    Without the hedge, a 20% fall would have cost Rs 2,10,000.
  - Below about 22,300 the over-hedge makes you NET POSITIVE
    in a crash. Pleasant, but it means the hedge is not
    neutral - it is a partial short.
  - Cost of 0.89% for 24 days. Annualised: 13.5%. You cannot
    run this permanently.

STEP 4B - PUT SPREAD VERSION (cheaper, caps the over-hedge)
___________________________________________________________

BUY  1 lot 24,000 PE @ 118.00   -Rs 8,850
SELL 1 lot 22,800 PE @  42.00   +Rs 3,150
NET DEBIT = 76.00 points        = Rs 5,700 (+ ~Rs 100 charges)

Max protection = (24,000 - 22,800 - 76) x 75
               = 1,124 x 75 = Rs 84,300

Index   Portfolio P&L   Spread payoff   Net (Rs)
______  _____________  ______________  __________
25,000             0         -5,800      -5,800
24,000       -42,000         -5,800     -47,800
23,000       -84,000        +69,200     -14,800
22,800       -92,400        +84,300      -8,100
22,000    -1,26,000         +84,300     -41,700
20,000    -2,10,000         +84,300    -1,25,700
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  Cheaper (Rs 5,800 vs Rs 8,910) and the protection is
  CAPPED below 22,800. In a true crash the simple put is
  far better; in a moderate correction the spread is more
  cost-efficient. Choose deliberately.

STEP 5 - WHAT YOU MUST ACCEPT
_____________________________

  - Most likely outcome: the index does not fall, you lose the
    full Rs 8,910, and the hedge "failed". It did not fail.
    It expired unused, like every insurance policy you have
    ever bought and not claimed on.
  - Basis risk: your portfolio is not the index. If your
    holdings fall while the index does not, the hedge pays
    nothing.
  - Lot granularity: you could not size this precisely.

Premiums, IV and charges illustrative. As at late 2026 -
verify at nseindia.com.
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Use case 2: covered calls against a holding you actually own

TEXT

SITUATION
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You own 1,000 shares of a hypothetical F&O-eligible company
bought at an average of Rs 1,620. Current price Rs 1,800.
Lot size 500, so you can write 2 lots.
You would be content to sell at Rs 1,900.
You have no intention of selling below Rs 1,900.

THE TRADE

SELL 2 lots 1,900 CE @ Rs 42.00, 25 days to expiry
Premium received = 42 x 500 x 2 = Rs 42,000
Charges (illustrative, 2 orders + STT on sale + GST)

STT on sale (0.1% of 42,000)	Rs 42.00
Brokerage 2 orders + GST	Rs 47.20
Exchange charges + GST	Rs 17.30
Total	Rs 106.50

NET PREMIUM = Rs 41,893.50

Return on the Rs 18,00,000 holding = 2.33% in 25 days

MARGIN

Because you own the shares, you can PLEDGE them as collateral, which covers the short-call margin after a haircut. You still need a cash component. Verify current rules with your broker.

OUTCOMES AT EXPIRY

Share at expiry	Shares' value (Rs)	Option outcome (Rs)	Total position value (Rs)
1,500	15,00,000	+41,894	15,41,894
1,700	17,00,000	+41,894	17,41,894
1,800	18,00,000	+41,894	18,41,894
1,900	19,00,000	+41,894	19,41,894
2,000	19,00,000*	+41,894	19,41,894
2,300	19,00,000*	+41,894	19,41,894

* shares are delivered at 1,900 via physical settlement

THE FOUR THINGS YOU MUST PLAN FOR
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1. PHYSICAL DELIVERY. If the share closes above 1,900, your 1,000 shares WILL be delivered. Keep them unencumbered and available. If they are pledged as collateral, unpledge them in time - pledge release takes a settlement cycle.

2. DELIVERY MARGIN RAMP. From about four sessions before

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   expiry, if the call is ITM you face escalating delivery
   margin. Ensure cash is available or close the position.

3. TAX EVENT. Delivery at 1,900 realises a capital gain of
   (1,900 - 1,620) x 1,000 = Rs 2,80,000, at a time of the
   market's choosing, not yours. The premium is taxed as
   business income (F&O income is business income in India);
   the share sale is a capital gain. Two different heads.
   Verify current treatment at incometax.gov.in.

4. THE OPPORTUNITY COST IS REAL. At Rs 2,300 you forgo
   Rs 4,00,000 of upside to have collected Rs 41,894.
   If you would regret that, do not write the call.

REPEATING IT
  Writing this every month is NOT a 2.33% monthly income
  stream. In a sustained rally you get assigned early in the
  move and miss the rest. In a decline the premium covers a
  fraction of the loss. The realistic long-run outcome of a
  systematic covered-call programme is a return pattern with
  LOWER volatility and LOWER expected return than simply
  holding - which may be exactly what you want, or not.
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Use case 3: a put spread instead of a stop loss

TEXT

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THE PROBLEM WITH STOP LOSSES IN INDIAN EQUITIES
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You hold a position and place a stop loss 8% below.
  - An overnight gap can open 12% below your stop. The stop
    becomes a market order at the gap price. You lose 12%,
    not 8%.
  - An intraday spike can trigger your stop and then reverse,
    taking you out of a position you still wanted.
  - Circuit limits can prevent execution entirely.

A PUT (OR PUT SPREAD) HAS NONE OF THESE PROBLEMS
  - It pays out regardless of the gap. A 12% gap down means
    the put is 12% in the money. No execution risk.
  - It cannot be "triggered" by an intraday wick.
  - Your maximum loss is known in advance, exactly.
  - BUT you pay a premium, and a stop loss is free.

WORKED COMPARISON (index version, Rs 10 lakh exposure)
──────────────────────────────────────────────────────

Index 25,000. Suppose you hold index exposure and want to
limit the damage from a fall below 24,000.

METHOD A: STOP LOSS AT 24,000
  Cost upfront:        Rs 0
  Loss if it drifts to 24,000:   -Rs 40,000 (-4%)
  Loss if it GAPS to 23,200:     -Rs 72,000 (-7.2%)
  Loss if it gaps to 22,000:   -Rs 1,20,000 (-12%)
  Risk of being stopped out on a wick and then rallying: HIGH

METHOD B: BUY THE 24,000 PE @ 118.00
  Cost upfront:        Rs 8,910 (0.89%)
  Loss at 24,000:      -Rs 40,000 - 8,910 = -Rs 48,910
  Loss at 23,200:      -Rs 80,000 + 60,000 - 8,910 = -Rs 28,910
  Loss at 22,000:    -Rs 1,20,000 + 1,50,000 - 8,910 = +Rs 21,090
  Risk of being stopped out on a wick: ZERO

METHOD C: BUY THE 24,000/22,800 PUT SPREAD @ 76.00
  Cost upfront:        Rs 5,800 (0.58%)
  Loss at 24,000:      -Rs 45,800
  Loss at 23,200:      -Rs 80,000 + 60,000 - 5,800 = -Rs 25,800
  Loss at 22,000:    -Rs 1,20,000 + 84,300 - 5,800 = -Rs 41,500

THE TRADE-OFF, STATED PLAINLY
  A stop loss costs nothing and fails exactly when you need it
  (a gap). A put costs 0.89% and works exactly when you need it.
  You are paying 0.89% to remove gap risk and wick risk.

  Whether that is worth it depends on how likely a gap is and
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  how much a gap would hurt you. For a leveraged position or a
  position you cannot afford to lose, it usually is. For a
  small long-term holding you were going to keep anyway, it
  usually is not - the right answer there is to not use a stop
  loss at all.

Premiums illustrative. As at late 2026.
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Use case 4: a defined-risk event trade

TEXT

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SITUATION
═══════
A hypothetical F&O-eligible company reports quarterly results
in six days. You have done the fundamental work (see the
stock-analysis guide) and you have a view that the market's
expectations are too low. You want to express that view with
RISK YOU CAN DEFINE AND AFFORD TO LOSE ENTIRELY.

Share price                     Rs 1,800
Lot size                        500
Implied move priced by the
  options market (from the ATM
  straddle)                     +/- 6.2%  (Rs 1,688 to 1,912)

FIRST, THE CRITICAL CHECK
  The straddle is pricing a 6.2% move. For a LONG option
  position to profit you need the move to EXCEED 6.2% in
  your direction. "I think it will go up" is not enough.
  "I think it will go up more than 6.2%" is the actual bet.

  If you cannot articulate why the market's 6.2% estimate is
  wrong, you do not have a trade. You have a hunch, and the
  option market has already sold you the hunch at fair value.

STRUCTURE A: BUY THE 1,900 CE @ Rs 54.00
  Cost = 54 x 500 = Rs 27,000
  Break-even = 1,954 (+8.6%)
  Max loss = Rs 27,000 (100% of outlay)
  IV is elevated pre-results; expect a post-results IV crush.

STRUCTURE B: BULL CALL SPREAD 1,850 / 1,950
  BUY  1,850 CE @ 78.00   -Rs 39,000
  SELL 1,950 CE @ 34.00   +Rs 17,000
  NET DEBIT              = Rs 22,000
  Max profit = (100 - 44) x 500 = Rs 28,000
  Break-even = 1,894 (+5.2%)
  Max loss = Rs 22,000

  The spread's short leg recovers part of the inflated
  pre-event premium, so it suffers LESS from IV crush.
  Break-even is also lower (5.2% vs 8.6%).

OUTCOME TABLE
Share at    Structure A      Structure B
results     (long 1,900 CE)  (1,850/1,950 spread)
________    _____________    ___________________

 1,600          -27,000             -22,000
 1,750          -27,000             -22,000
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 1,850          -27,000             -22,000
 1,894          -27,000                   0
 1,900          -27,000              +3,000
 1,954                0             +28,000
 2,000          +23,000             +28,000
 2,100          +73,000             +28,000
 2,300        +1,73,000             +28,000

  A wins big on a very large move. B wins on a moderate move
  and loses less when wrong. Neither wins if the share rises
  5% - the move everyone would call "a good result".

THE SIZING RULE FOR EVENT TRADES
  Whatever you put in, assume it is gone. An event trade is
  binary and you have no information advantage over the
  institutions that cover the company. Size it at a level
  where total loss is an irritation, not a problem - for most
  people that is well under 1% of net worth, not 10% of the
  trading account.

AND THE HONEST CAVEAT
  Trading individual company results with options is close to
  gambling for anyone without a genuine informational or
  analytical edge. Having read the annual report is not an
  edge; so has everyone else. This use case is "defensible"
  only in the narrow sense that the risk is defined. The
  expected value is still negative for most participants.

Premiums illustrative. As at late 2026.
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If you still want to trade F&O: the honest prerequisites

You have read the evidence and you still want to proceed. That is your decision to make. What follows is the minimum standard a serious person would hold themselves to — not a path to profit, but a set of conditions below which the activity is indefensible.

Before anything else

If you do not have an emergency fund, health insurance, term life cover (if anyone depends on you), zero high-interest debt, and a funded long-term investment plan, you should not be trading derivatives. Derivatives are the last thing you add to a financial life, not the first. See the investing foundations guide.

The ten prerequisites

1. **Risk capital you can lose entirely**
   Not savings. Not borrowed money. Not money with a purpose. A sum whose complete loss changes nothing material in your life. For most salaried software engineers in India that is a small number, and it should be.

2. **A fully funded long-term portfolio first**
   Index funds, EPF/NPS, emergency fund — all in place and on autopilot. Derivatives come out of the discretionary surplus after that, never instead of it.

3. **Time you actually have**
   Intraday derivatives trading is incompatible with a full-time engineering job. Not "difficult" — incompatible. If your plan requires watching a chart during standups, the plan is already broken.

4. **A tested edge, written down**
   A specific, falsifiable hypothesis about why you expect positive expectancy, tested over a meaningful sample. "I read charts well" is not an edge. See the trading strategy guide for how to specify and test one.

5. **Defined risk on every single position**
   Maximum loss computed and acceptable *before* entry. In practice this means long options and debit/credit spreads with bought wings. No naked short options, ever.

6. **No naked short options. Ever.**
   This is not a preference. For a retail account with no vega hedge, no intraday risk system and no ability to trade through a gap, an uncovered short option is an unmanaged tail exposure. The high win rate is the bait.

7. **Position sizing by exposure, not by premium**
   Compute delta-equivalent notional. Cap total portfolio delta exposure as a percentage of net worth, not of the trading account.

8. **A written trading plan**
   Entry criteria, exit criteria, stop rule, size rule, maximum positions, maximum daily loss, what you will do on a gap. Written before the session, not during it.

9. **Six months of paper trading, minimum**
   With real position sizes, real cost modelling, a real journal, and real honesty about fills you would not have got. If you cannot produce a positive expectancy on paper over six months and 100+ trades, you do not have an edge — and paper results are systematically better than live results.

10 **A pre-committed stop condition**

A drawdown level at which you stop entirely and go back to investing. Write it down now, while you are calm. −25% of the trading capital is a reasonable place to put it. You will want to move it. That is exactly why you write it down in advance.

⚠ **Paper trading overstates your results**

Paper trading has no slippage, perfect fills at mid, no emotional pressure, no margin calls, and a retrospective view of what "you would have done". Expect your live results to be materially worse than your paper results. If your paper expectancy is marginally positive, your live expectancy is almost certainly negative.

The things that disqualify you, today

- ☐ You are trading with borrowed money, a personal loan, or a credit card.
- ☐ You have pledged your long-term equity portfolio to fund derivatives margin.
- ☐ You cannot state your current position's net delta, theta and vega in rupees.
- ☐ You have ever increased position size specifically to recover a loss.
- ☐ You are following tips, a Telegram group, or a paid signal service.
- ☐ You are trading primarily on expiry day.
- ☐ You hold uncovered short options.
- ☐ Your trading affects your sleep, your work, or your relationships.
- ☐ You have not computed your round-trip transaction cost as a percentage of capital at risk.
- ☐ You do not have a written plan you could show to someone else.

Glossary

Term	Definition
ATM / ITM / OTM	At / in / out of the money. Describes strike relative to spot.
Assignment	Being required, as an option writer, to fulfil the contract when the buyer exercises.
Auction settlement	Exchange procurement of shares you failed to deliver, at your cost plus penalty.
Basis	Futures price minus spot price. Converges to zero at expiry.
Backwardation	Futures trading below spot (negative basis).
Contango	Futures trading above spot (positive basis). Normal for equity indices.
CE / PE	Call European / Put European — the Indian contract notation.
Cost of carry	Interest forgone minus dividends received, from holding the underlying to expiry.
Delivery margin	Escalating margin in the expiry week on stock F&O positions heading to physical settlement.
DNE	Do Not Exercise — an instruction to forgo exercise of a marginally ITM option. Availability varies; verify.
ELM	Extreme Loss Margin — an additional margin layer above SPAN.
Exercise	Invoking the right conferred by an option.
Greeks	Delta, gamma, theta, vega, rho — sensitivities of option value to its inputs.
Haircut	The discount applied to pledged securities when computing collateral value.
Implied volatility	The volatility input that makes a pricing model output the observed market premium.
India VIX	NSE's volatility index derived from near-term index option prices.
Intrinsic value	$\max(S-K,0)$ for a call, $\max(K-S,0)$ for a put. The exercise value right now.
IV crush	A sharp fall in implied volatility after an event resolves, damaging long-option positions.

Lot size	Units of underlying per contract. Set by the exchange; revised periodically.
Mark to market	Daily cash settlement of futures gains and losses.
Moneyness	Relationship of strike to spot.
Notional value	Lot size × underlying price. Your real exposure.
Open interest	Contracts outstanding and not yet closed.
Peak margin	The highest margin requirement observed in intraday snapshots.
Physical settlement	Settlement by actual delivery of shares. Applies to Indian single-stock F&O.
Pin risk	Uncertainty about exercise when the underlying closes very near a strike.
Premium	The price of an option, quoted per unit of underlying.
Put–call parity	The no-arbitrage relationship linking call, put, spot and discounted strike.
Roll	Closing a near-expiry position and opening the equivalent in a later expiry.
SPAN	Standard Portfolio Analysis of Risk — the scenario-based margin system.
Skew / smile	The pattern of implied volatility across strikes.
Spot	Current market price of the underlying.
Square off	Close a position by taking the opposite trade.
STT	Securities Transaction Tax. Charged on premium for option sales, on settlement value for exercise.
Strike	The fixed price in an option contract.
Time value	Premium minus intrinsic value. Zero at expiry.
Vertical spread	Two options, same expiry, different strikes.
Write	To sell an option you do not already hold.

Frequently Asked Questions

Is it really true that about 9 out of 10 F&O traders lose money?

That is what SEBI's studies of actual participant data report. The September 2024 study covering FY22–FY24 found roughly 93% of individual F&O traders made losses, with aggregate individual net losses in the region of ₹1.8 lakh crore over the period. A 2025 follow-up covering FY25 found roughly 91% of individual index-option traders losing. Treat the precise figures as approximate and verify the latest at sebi.gov.in — SEBI revises the methodology and the period. The structural conclusion is stable across every version: derivatives are a negative-sum game after costs and most individuals lose.

If the odds are that bad, why does anyone trade F&O?

Several reasons. Hedgers and institutions use it for its intended purpose — transferring risk — and accept a negative expected P&L in exchange for reduced variance. Professional firms have genuine structural edges (market making, latency, scale, cost structure) and profit consistently. Individuals are attracted by leverage, the accessibility of the product, visible survivorship-biased success stories, and the fact that a high-win-rate strategy feels like skill for a long time before it fails. The last group is the one SEBI's statistics are measuring.

Can I start F&O with ₹50,000?

After the 2024–25 minimum contract size increases, one index option lot often requires several thousand rupees of premium and one futures or short-option lot requires ₹2 lakh or more of margin. With ₹50,000 you are restricted to buying one or two lots of options, which means a single position is 15–30% of your capital — a position size that guarantees ruin over a sample of trades. The honest answer is that ₹50,000 is not enough capital to size derivative positions responsibly, and the correct use of ₹50,000 for almost everyone is a low-cost index fund. Verify current lot sizes and margin requirements at nseindia.com.

Is option buying safer than option selling because the loss is capped?

The maximum loss is capped, which is a genuine advantage. The probability of incurring that maximum loss is very high — for an out-of-the-money option it is typically 60–90%. Option buying has a low win rate with a bounded loss; option selling has a high win rate with an unbounded loss. Both have negative expectancy for a participant without an edge; they fail in different ways. Buying fails through a slow grind of 100% losses; selling fails through one catastrophic event after a long winning run. Buying is survivable; selling can end your financial life. In that narrow sense, yes, buying is "safer".

Everyone says option sellers win because most options expire worthless. Is that right?

The statistic is often misquoted. A commonly cited figure is that a large share of options expire worthless, but a substantial proportion of positions are closed before expiry rather than held to it, so "expire worthless" is not the same as "sellers profit". More importantly, the statistic is irrelevant to expectancy. A seller can win 90% of the time and still lose money, because the losses are many times

the size of the wins. The worked short-strangle expectancy calculation in this guide shows a negative expected value despite an 86% win probability. Win rate without loss magnitude is a meaningless number.

What is the single most dangerous thing a beginner can do in F&O?

Sell uncovered (naked) options, particularly short straddles and strangles, and particularly on expiry day. It combines a high win rate (which builds false confidence and encourages size escalation), unbounded loss, vega exposure that hurts you on a volatility spike with no price move, margin that expands exactly when the position is losing, and gap risk that no stop loss can protect against. The second most dangerous thing is buying cheap out-of-the-money options on expiry day, which has a deeply negative expected value and a 10–20% cost ratio.

I was right about the direction but still lost money. How?

Three common causes, usually in combination. First, the move did not exceed your break-even — a long call needs the index to rise by more than the premium paid. Second, implied volatility fell (IV crush), which reduces the premium regardless of direction; this is routine after a scheduled event. Third, time passed (theta), which erodes premium every day. The worked example in the volatility section shows a correct directional call losing 35% because the vega and theta losses exceeded the delta gain. Buying an option is a bet on direction *and* magnitude *and* speed *and* volatility, not direction alone.

What exactly is the STT trap on exercised options?

STT on a *sold* option is charged on the premium, which is small. STT on an option that is *exercised* at expiry is charged on the settlement (intrinsic) value, which for a deep in-the-money option can be many times the premium you paid. A position with a small profit can become a loss purely from this. The worked example in the costs section shows a ₹750 gross profit turning into a ₹1,125 net loss. The rule: sell your in-the-money options in the market before expiry rather than letting them be exercised, unless you have computed that exercise is cheaper. Verify current STT rates and bases at incometax.gov.in.

What happens if I hold a stock option into expiry and it is in the money?

It is automatically exercised and settles by **physical delivery**. Depending on which of the four positions you hold, you will either need to produce the full cash value of the contract (lakhs of rupees, not your margin) or produce the actual shares in your demat account. If you cannot deliver shares you owe, the exchange buys them in an auction at your cost plus penalty, which can be far worse than the market price. From roughly four sessions before expiry you also face escalating delivery margin. The safe rule is to close all single-stock F&O positions before expiry week unless you specifically intend to settle physically and have the cash or shares ready.

Did SEBI's 2024–25 curbs fix the problem?

Partially. Weekly expiries were limited to one benchmark index per exchange, minimum contract sizes were raised substantially, option premium must be collected upfront from buyers, the calendar-spread

margin benefit was removed on expiry day, position limits are monitored intraday, and an additional tail-risk margin applies to short options on expiry day. Short-dated option volumes and the number of unique individual participants fell. But SEBI's 2025 follow-up found roughly 91% of individual index-option traders still losing — the loss *rate* barely moved. The measures reduced access and leverage; they did not change the zero-sum structure, the cost stack or the skill gap. Verify the current framework at sebi.gov.in.

How are F&O profits and losses taxed in India?

Income from exchange-traded equity derivatives is generally treated as **non-speculative business income** under the Income-tax Act, taxed at your slab rate, with losses able to be set off against other business income and carried forward (subject to conditions). This is different from capital gains treatment on shares. Expenses including brokerage, internet, and depreciation on equipment may be deductible. Tax audit requirements may apply above certain turnover thresholds, and the definition of "turnover" for F&O is specific. Advance tax obligations apply. This is a genuinely complex area — **as at FY 2026–27, verify at incometax.gov.in** and engage a chartered accountant. Note also that the Income-tax Act 2025 renumbered and restructured provisions from 1 April 2026.

Should I use a stop loss on option positions?

A stop loss on a long option is of limited use because the instrument already has a defined maximum loss and because option prices gap and spike on thin liquidity, triggering stops on noise. A stop loss on a short option is close to useless in the scenario that matters — an overnight gap bypasses it entirely, and that is precisely the event that destroys short-option positions. The better risk controls for options are position sizing by delta-equivalent exposure, buying protective wings so the maximum loss is structurally capped, and simply not holding positions whose worst case you cannot absorb.

Can I hedge my ₹5 lakh portfolio with index options?

With difficulty. After the minimum contract size increases, one index option lot typically covers ₹15–20 lakh of notional. Buying one put to hedge ₹5 lakh over-hedges by 3–4x, which converts your long portfolio into a net short position in a crash — that is a new bet, not a hedge. For portfolios below roughly ₹15–20 lakh, the practical hedging tools are reducing equity exposure, holding more cash or short-duration debt, or simply accepting the volatility if your horizon is long. Verify current lot sizes at nseindia.com.

What is a realistic return expectation if I am good at this?

The honest answer is that the question is poorly framed. Professional systematic traders with genuine edges, infrastructure and risk teams target returns measured in the tens of percent with significant drawdowns, and many fail. For a retail participant, the realistic expectation before developing a demonstrated edge is a loss. If you are determined to have a number: a plausible outcome for a disciplined, well-capitalised retail trader with a genuine small edge is to roughly match a broad index after costs, with far more work, far more stress and far more tax friction. That is the *good* outcome, and it is a strong argument for simply buying the index.

> **Is paper trading actually useful, or is it a waste of time?**
>
> It is useful for exactly one thing: discovering whether your strategy has positive expectancy before you pay to find out. It is not useful for developing psychology, because there is no money at stake. It also systematically overstates results — no slippage, perfect fills, no margin calls, no emotional pressure, and hindsight contaminating your "would have done" decisions. Model costs aggressively, assume you get the worse side of every spread, keep a real journal, and treat a marginally positive paper expectancy as equivalent to a negative live one. Six months and 100+ trades is a reasonable minimum before risking money.

Decision checklist and strategy reference

Two tools to close with. First, an honest test of whether you should be in this market at all. Second, a one-page reference for the structures covered.

Should I trade derivatives at all?

Answer each honestly. Every "no" in Part A is disqualifying on its own.

PART A — FINANCIAL PRECONDITIONS (ALL MUST BE "YES")

- ☐ I have 6–12 months of expenses in an emergency fund, untouched.
- ☐ I have health insurance and, if anyone depends on me, term life cover.
- ☐ I have no outstanding high-interest debt (credit cards, personal loans).
- ☐ I have a funded, automated, long-term investment plan already running.
- ☐ The capital I would trade with is money whose total loss changes nothing material in my life.
- ☐ None of that capital is borrowed, and none of my long-term portfolio is pledged as margin for it.
- ☐ I can afford one full contract lot and still size positions at a small fraction of my trading capital.

PART B — COMPETENCE PRECONDITIONS (ALL MUST BE "YES")

- ☐ I can compute intrinsic value, time value and break-even for any option without looking it up.
- ☐ I can state my position's net delta, gamma, theta and vega, converted into rupees.

- ☐ I can compute the expected 1-standard-deviation move from implied volatility and compare it to my break-even.
- ☐ I know my round-trip transaction cost as a percentage of the capital at risk, for the specific contracts I trade.
- ☐ I understand physical settlement and I have a calendar process for expiry week.
- ☐ I have a written strategy specification with entry, exit, stop and size rules.
- ☐ I have paper-traded that specification for at least six months and 100 trades with realistic cost modelling.
- ☐ I have a pre-written drawdown level at which I stop completely.

PART C — THE DISQUALIFIERS (ANY "YES" MEANS STOP)

- ☐ I would hold uncovered short options.
- ☐ I would trade primarily on expiry day.
- ☐ I follow tips, signal groups or paid "calls".
- ☐ I have increased position size to recover a loss.
- ☐ I expect derivatives to replace or supplement my income within a year.
- ☐ My trading affects my sleep, my work or my relationships.
- ☐ I cannot articulate, in one sentence, what edge I have and who is on the other side of my trade.

✓ **The most defensible outcome of this checklist**

For the large majority of Indian software engineers reading this, the correct answer is: do not trade derivatives. Use the time you would have spent on it to increase your income and your savings rate, and put the money in a low-cost index fund. That is not a cop-out — it is what the evidence supports, and it is what most people who have been through a full cycle in derivatives eventually conclude.

Strategy quick reference

If your view is…	…and you want defined risk	…and you accept undefined risk	Key hidden risk
Strongly up	Long call, or bull call spread for a bounded target	Long futures	Time decay and IV crush on the long call; MTM and gap risk on futures
Mildly up / sideways, and you own the stock	Covered call	—	Capped upside; physical delivery; you still own all the downside
Mildly up / sideways, and you want the stock lower	Cash-secured put (fully collateralised)	Naked short put	You buy precisely when the news is bad; large left tail
Strongly down	Long put, or bear put spread	Short futures	You pay the skew premium; indices drift up over time
Protect a holding for a defined window	Protective put, or collar	—	Cost drag if run permanently; lot-size granularity; collar caps upside
Big move, direction unknown	Long straddle or strangle	—	Double theta and double vega; needs the move to exceed what is priced
No move expected	Iron condor (wings bought)	Short straddle / strangle — **not suitable for retail**	Condor needs a >90% win rate after costs; naked versions have unbounded loss
Pin at a specific level	Butterfly	—	Low probability; eight fills round trip; only pays at expiry
Flat now, volatility later	Calendar spread	—	Expiry-day margin benefit removed; far-month liquidity; stock-settlement mismatch

No view	**No position.** This is a valid and frequently optimal choice.	—	The urge to have a position is not a view

A pre-trade routine, if you proceed

1. **State the view in one sentence**
   Including magnitude and timeframe. "Up" is not a view; "up more than 2% within 12 trading days" is.

2. **Compute the implied move**
   Spot × IV × √t. Compare your target to it. If the market already prices your move, you have no trade.

3. **Select the structure**
   From the reference table above. Prefer defined risk. Never uncovered short options.

4. **Compute max loss in rupees**
   Not percentage. The actual number. Then ask whether you would be comfortable if that happened today.

5. **Compute delta-equivalent exposure**
   Lots × lot size × delta × spot. Check it against your net worth, not your trading account.

6. **Compute the round-trip cost**
   As a percentage of capital at risk. If it exceeds 2–3%, reconsider the contract.

7. **Check the margin requirement**
   On the exchange calculator. Read it as a risk opinion, not an obstacle.

8. **Write the exit plan**
   Profit target, loss limit, time stop, and what you do on a gap — before entering.

9. **Check the expiry calendar**
   Especially for single stocks. Set a reminder three sessions before expiry.

10. **Log the trade**
    Pre-trade entry in your journal, with the hypothesis. See the trading strategy guide for a full template.

Closing note

This guide has given you the mechanics, the arithmetic and the risk structure of Indian equity derivatives. It has deliberately not given you a strategy, because no honest guide can. The evidence from SEBI's own participant data is that the overwhelming majority of individuals who use these instruments lose money, and that the losses are large in aggregate. Nothing in this guide changes that. Treat derivatives as what they are — a professional risk-transfer tool with narrow legitimate retail applications — and the knowledge here will serve you. Treat them as a way to make money quickly and the knowledge here will only make you a more articulate participant in the 90%. Every figure, rate, specification and regulation above is illustrative and dated to late 2026; verify current details at **nseindia.com**, **bseindia.com**, **sebi.gov.in** and **incometax.gov.in**, and consult a SEBI-registered Investment Adviser for personal advice.